

Tutorial 6

Statistical Computation and Analysis
Spring 2026

Tutorial Outline

- Short revision: Type I & II errors and z-test
- Continuous data
- Normal models
- Bayesian p-values
- Paired data
- Unpaired data
- Shared versus separate sigma in Bayesian models
- Student-t distribution & Robust models
- Frequentist t-tests

Type I and Type II Errors

- In a hypothesis test, there are two possible states of reality:

H_0 " is true"

- or

H_0 " is false"

- The test has two possible decisions:

Reject H_0

- or

Fail to reject H_0

Type I and Type II Errors

- **Type I error** - rejecting the null hypothesis when the null hypothesis is true.
- **Type II error** - failing to reject a false null model

		Reality	
		H_0	H_1
Hypothesis testing decision	H_0	True Posetive	Type 2 error (false Negative)
	H_1	Type 1 error (False Posetive)	True Negative

Significance Level and p-value

- The decision uses a significance threshold α .
- It defines the long-run Type I error rate allowed by the test decision rule.
- Usually:

$$\alpha = 0.05$$

Significance Level and p-value

- The **p-value** is calculated from the observed data.
- The probability of obtaining a test statistic at least as extreme as the one we observed, assuming that the null hypothesis is true.
- If $p < \alpha$ we reject the null hypothesis

Power and Type II error

- A Type II error happens when:

H_0 is false but the test decision is: Fail to reject H_0

- The probability of a Type II error is denoted by:

$$\beta$$

- Statistical power is the probability of detecting a real effect:

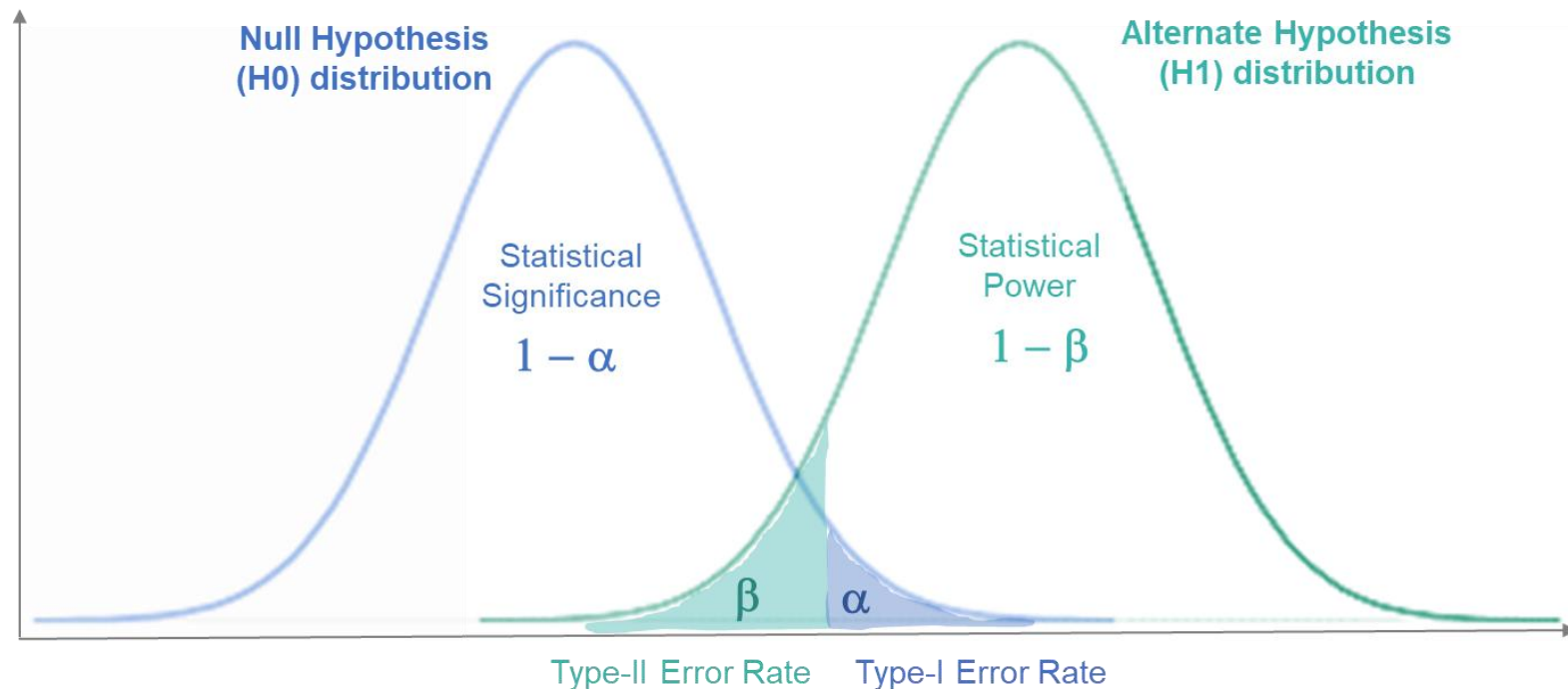
$$Power = 1 - \beta$$

- If power is 0.80, this means:

- If the assumed effect really exists, then in repeated experiments the test will reject H_0 about 80% of the time.

Power and Type II error

- Power increases when:
 - the effect size is larger
 - the sample size is larger
 - the noise / variance is smaller
 - α is larger
- Connection:
 - $\alpha \rightarrow$ Type I error
 - $\beta \rightarrow$ Type II error
 - $1 - \beta \rightarrow$ Power



Frequentist Two-proportion Z-test

- Frequentist hypothesis tests use a statistic whose sampling distribution is known.
- Z-test use the z statistic, $z \sim N(0, 1)$
- Under the null hypothesis we test whether calculating the statistic on our data is consistent with the null sampling distribution.

Frequentist Two-proportion Z-test

- **Assumptions**

1. Random sampling
2. Independent groups
3. Large sample size

The test relies on a normal approximation. If the sample is too small, the approximation may be unreliable.

If the assumptions are not met, use a different test or interpret the result cautiously.

Frequentist Two-proportion z-Test

- Null hypothesis is:

$$H_0: \theta_A = \theta_B$$

- The two-sided alternative is:

$$H_1: \theta_A \neq \theta_B$$

Z Statistic

$$z = \frac{\textit{observed estimate} - \textit{null value}}{\textit{standard error}}$$

$$z = \frac{\hat{\theta} - \theta_0}{SE}$$

Z Statistic - Two-proportion

$$Z = \frac{\hat{\theta}_A - \hat{\theta}_B}{\sqrt{\hat{\theta}_{pooled}(1 - \hat{\theta}_{pooled}) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}}$$

Where:

$$\hat{\theta}_{pooled} = \frac{z_A + z_B}{n_A + n_B}$$

Z Statistic - Interpretation

- **A z value tells us how many standard errors the observed estimate is away from the null value.**
- For example:

$$z=2$$

- Means that the observed estimate is two standard errors above what we would expect under the null model.

$$z=-2$$

- Means that the observed estimate is two standard errors below the null value.

Two-tailed z-test

- We ask whether the effect is different from zero in either direction.
- If the significance level is:

$$\alpha = 0.05$$

- And we split it between the two tails:

$$\frac{\alpha}{2} = 0.025$$

- The critical values are approximately:

$$z = -1.96, z = 1.96$$

- If the z statistic we calculated using our data is:

$$|z_{\text{observed}}| > 1.96$$

- The observed data is considered extreme under the null model.

Two-sided p-Value

- **p-Value**

- The probability of obtaining a test statistic at least as extreme as the one we observed, assuming that the null hypothesis is true.

- A two-sided test –

$$p = 2P(Z \geq |z_{\text{observed}}|)$$

- If

$$p < \alpha$$

- we reject the null hypothesis.

z-Test

```
theta_hat_A = z_A / n_A  
theta_hat_B = z_B / n_B
```

```
frequentist_difference = theta_hat_A - theta_hat_B
```

```
# Under H0, both groups share one pooled probability.  
pooled_null_rate = (z_A + z_B) / (n_A + n_B)  
  
standard_error_null = np.sqrt(  
    pooled_null_rate  
    * (1 - pooled_null_rate)  
    * (1 / n_A + 1 / n_B)  
)
```

```
z_statistic = frequentist_difference / standard_error_null  
two_sided_p_value = 2 * stats.norm.sf(abs(z_statistic))  
  
z_critical = stats.norm.ppf(1 - SIGNIFICANCE_ALPHA / 2)
```

```
ci_low = frequentist_difference - z_critical * standard_error_null  
ci_high = frequentist_difference + z_critical * standard_error_null
```

Calculating the thetas

Calculating the pooled
theta and the std under the
null model

Calculating the z statistic,
p-value and z critical value

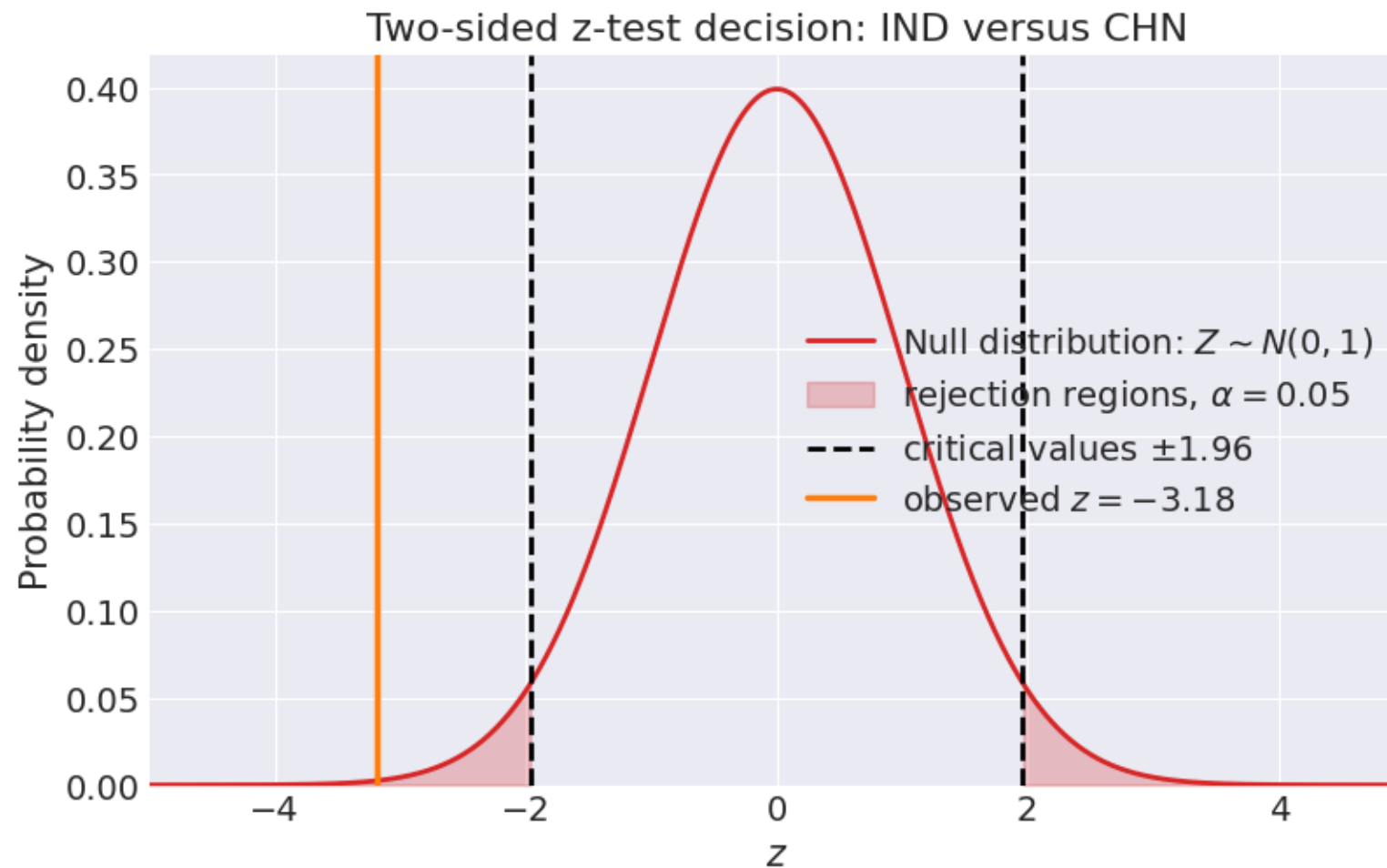
Calculating the confidence
interval

z-Test

	comparison	IND n	IND biotech	IND observed rate	CHN n	CHN biotech	CHN observed rate	observed difference
0	IND - CHN	1596	22	0.0138	1544	47	0.0304	-0.0167

pooled null rate	standard error under null	z statistic	two-sided p-value	95% normal CI low	95% normal CI high	reject at alpha=0.05
0.0220	0.0052	-3.18	0.001459	-0.0269	-0.0064	True

z-Test



Statistical reporting

z-test interpretation:

- Observed $z = -3.18$
- Critical values = ± 1.96
- Two-sided p-value = 0.001459
- Decision: Reject H_0 at $\alpha = 0.05$.

Confidence interval interpretation:

- Observed difference = -0.0167
- 95% CI = [-0.0270, -0.0064]
- The CI does not include 0, consistent with rejecting H_0 .
- A two-sided two-proportion z-test showed a statistically significant difference between the two countries, $z = -3.18$, $p = 0.001459$. The observed difference was -0.0167 , with a 95% confidence interval of $[-0.0270, -0.0064]$. Because the interval does not include 0, the confidence interval is consistent with rejecting the null hypothesis at $\alpha = 0.05$.

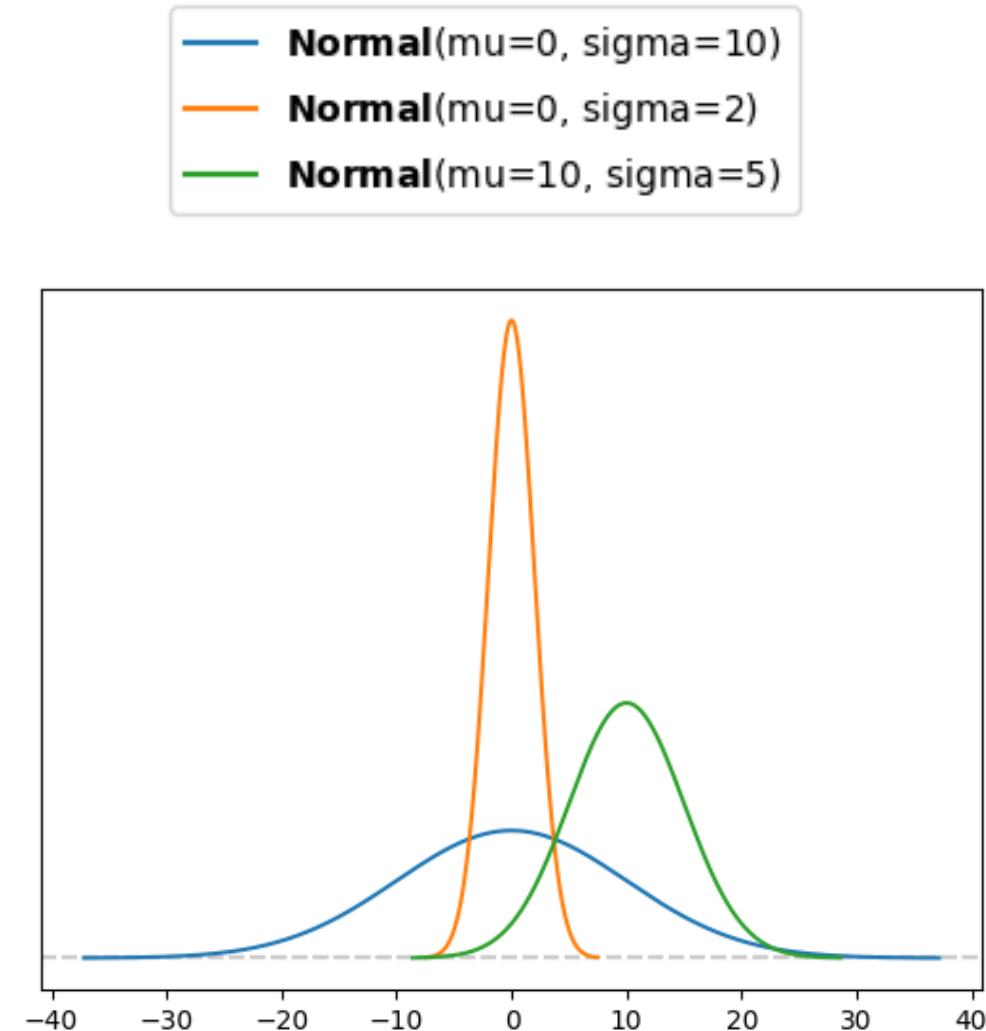
Normal Models

Continuous data

- Continuous measurements have units
- Require a continuous likelihood
- Examples:
 1. Weight, measured in kg
 2. Height, measured in cm
 3. Reaction time, measured in milliseconds
 4. Blood pressure, measured in mmHg

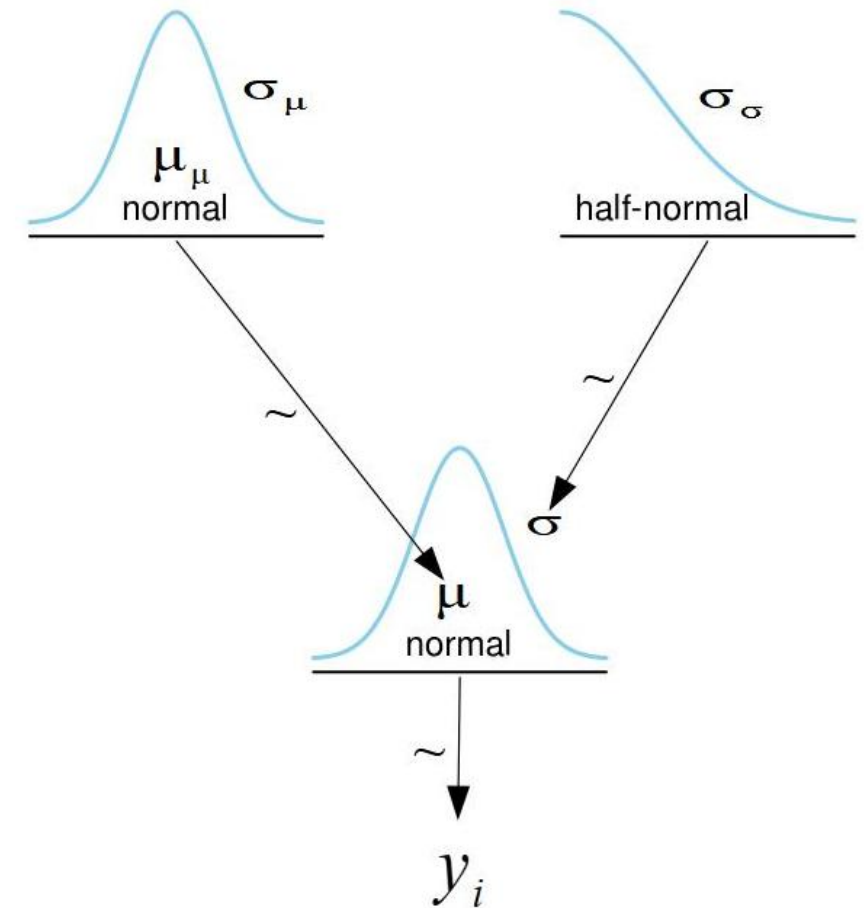
Normal Distribution

- Characterized by two parameters:
 - μ - center
 - σ - spread
- If we model our data using a normal distribution, we will have a posterior distribution for each parameter.



Normal Distribution

- Each parameter needs its own prior
 - Two priors
- We will use a normal prior for μ
- And a half normal prior for σ
 - Cannot be negative



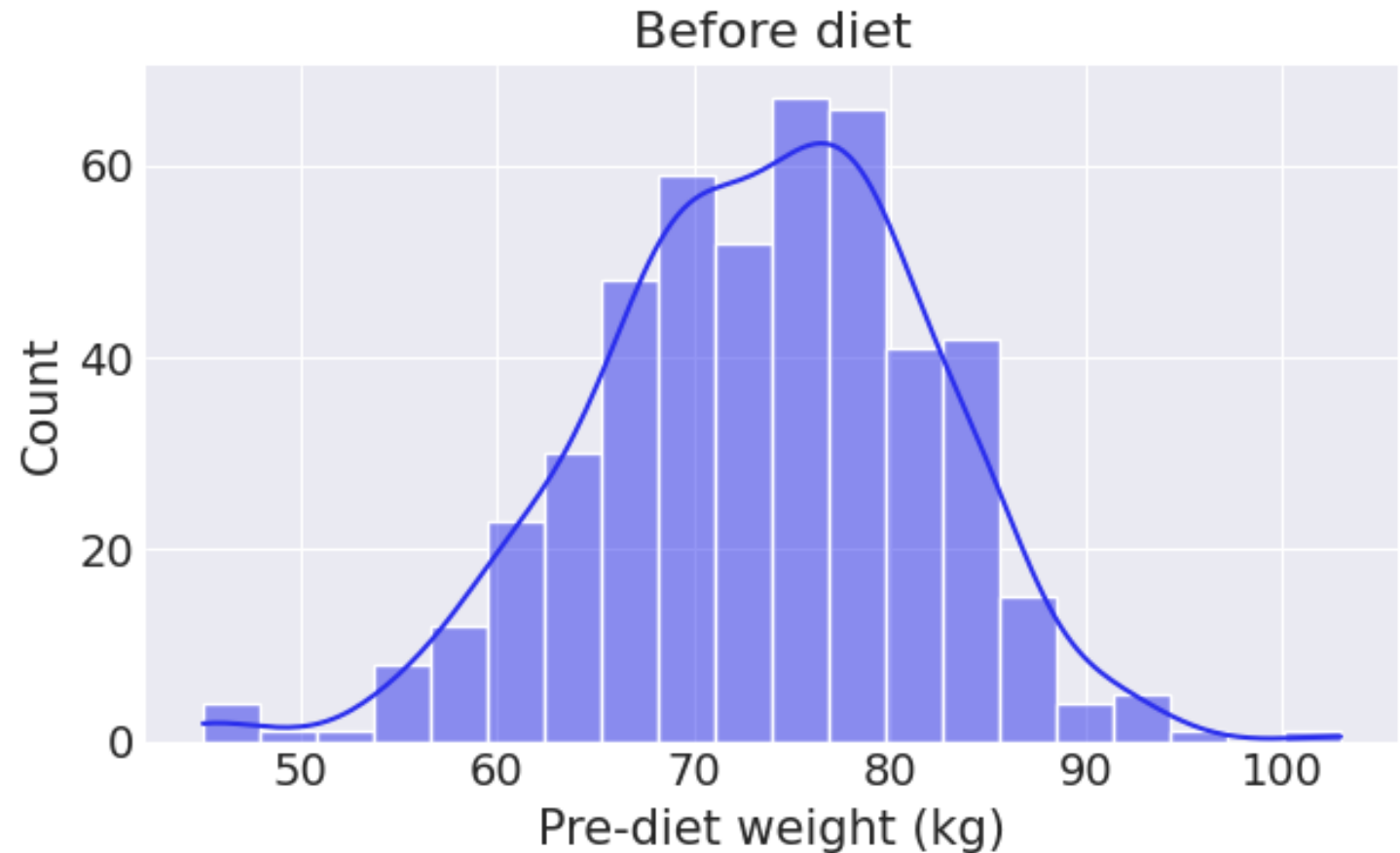
Normal Distribution Bayesian Model

- The generated dataset contains measurements from the same people before and after six weeks of dieting:
 - pre.weight: weight in kg before the diet
 - weight6weeks: weight in kg after six weeks

	Person	gender	Age	Height	pre.weight	Diet	weight6weeks	gender_label	change
0	1	0	22	159	58.0	1	54.2	female	-3.8
1	2	0	46	192	60.0	1	54.0	female	-6.0
2	3	0	55	170	64.0	1	63.3	female	-0.7
3	4	0	33	171	64.0	1	61.1	female	-2.9
4	5	0	50	170	65.0	1	62.2	female	-2.8

Normal Distribution Bayesian Model

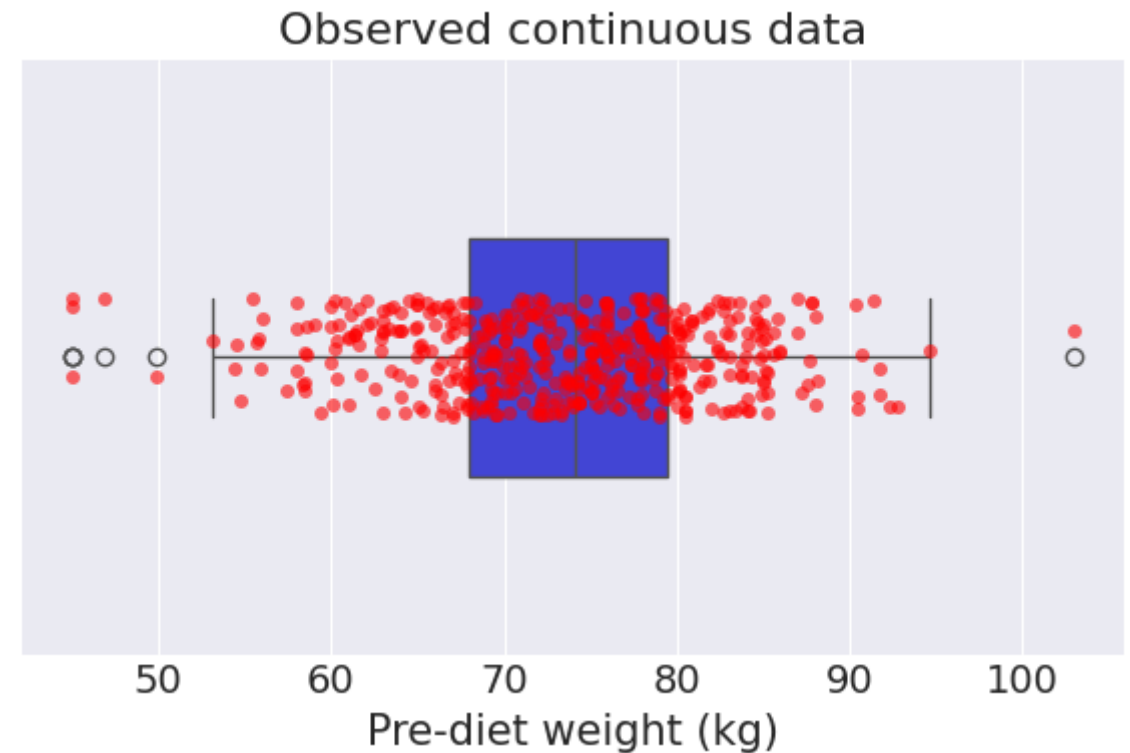
- Let's model the pre-diet weight data



Normal Distribution Bayesian Model

- **Box plot –**

- Line inside the box: median
- The box: quartiles (25th and 75th percentiles)
- Whiskers: Range excluding outliers
- Dots: outliers (usually more than 1.5 IQR beyond IQR)



Normal Distribution Bayesian Model

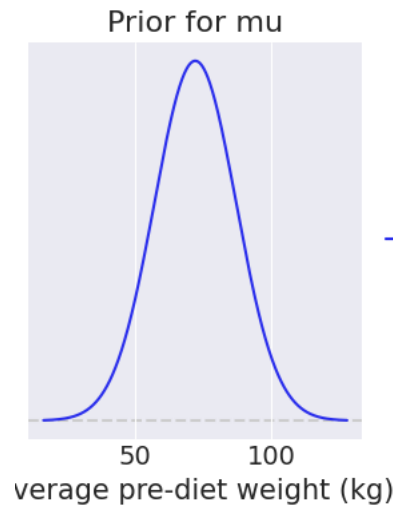
- **Choosing priors**

- Weight is measured in kilograms.
- Therefore, our priors will be chosen in kilograms

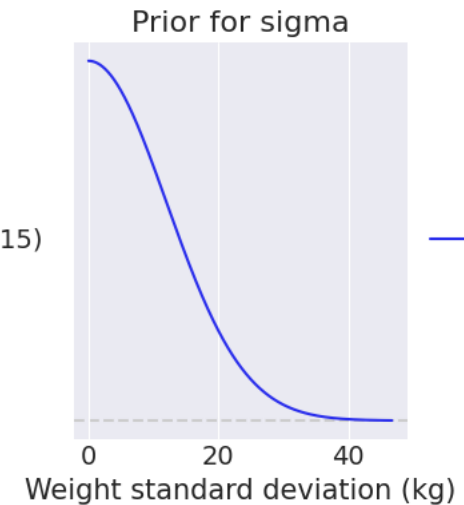
$$y \sim \text{Normal}(\mu, \sigma)$$

$$\mu \sim \text{Normal}(72, 15)$$

$$\sigma \sim \text{HalfNormal}(12)$$



— Normal(mu=72, sigma=15)



— HalfNormal(sigma=12)

Normal Distribution Bayesian Model

- Model

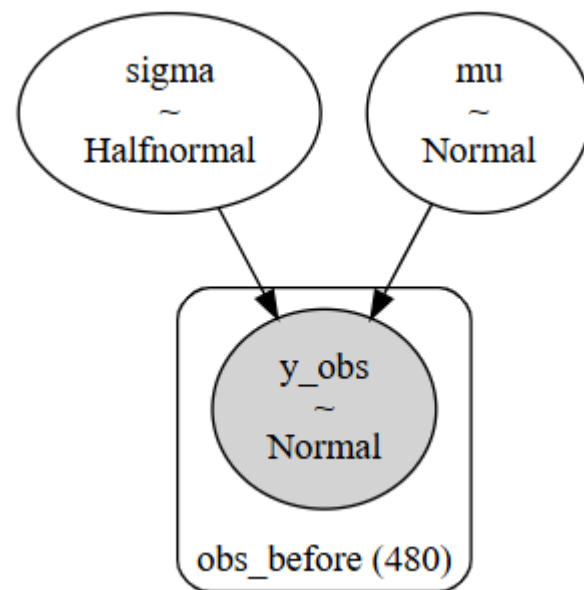
```
coords_food_pre = {"obs_before": np.arange(len(before))}

with pm.Model(coords=coords_food_pre) as normal_before_model:

    mu = pm.Normal("mu", mu=mu_prior_mean, sigma=mu_prior_sigma)
    sigma = pm.HalfNormal("sigma", sigma=sigma_prior_sigma)

    y_obs = pm.Normal("y_obs", mu=mu, sigma=sigma, observed=before, dims="obs_before")

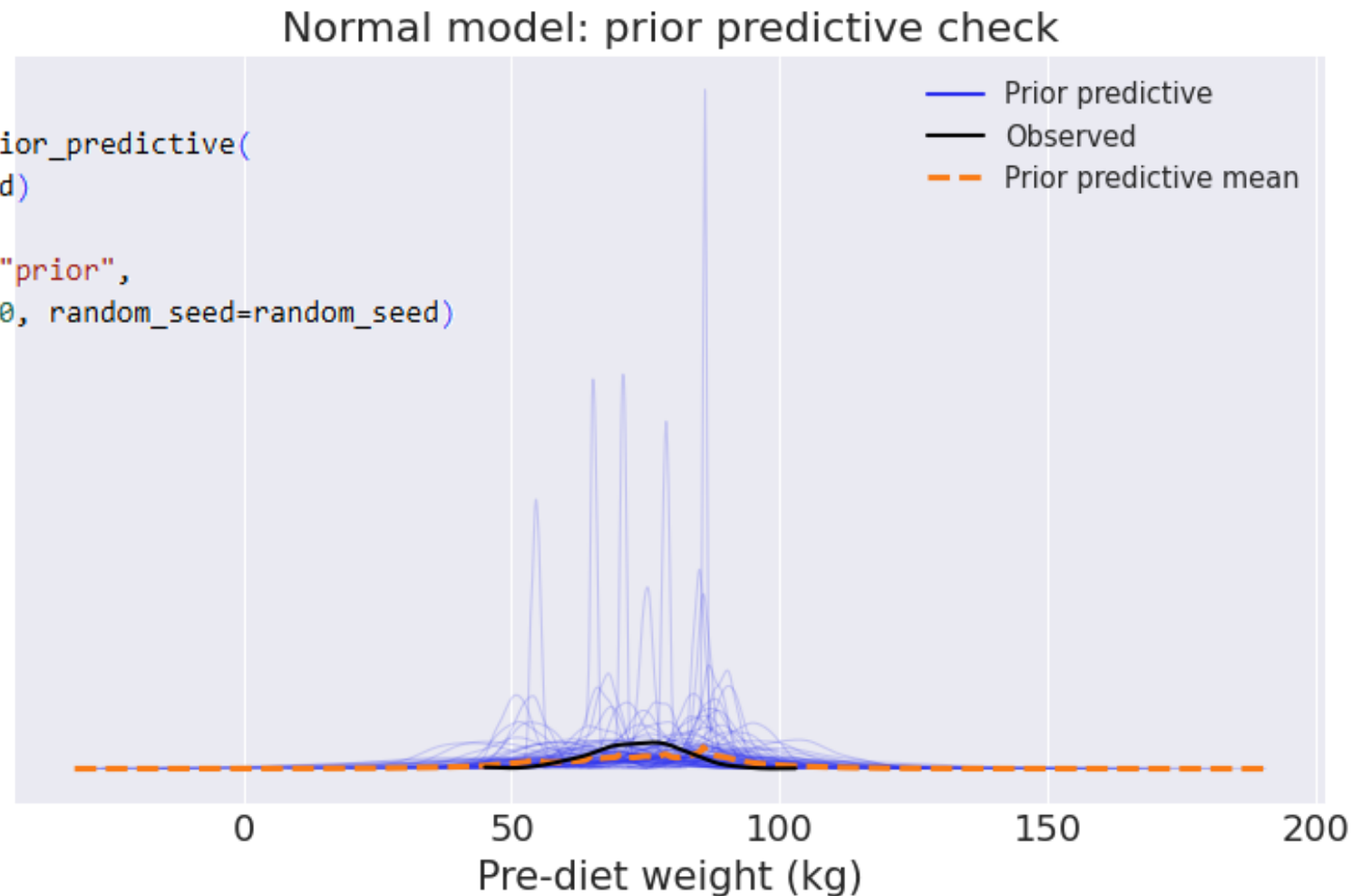
pm.model_to_graphviz(normal_before_model)
```



Normal Distribution Bayesian Model

- Prior predictive check

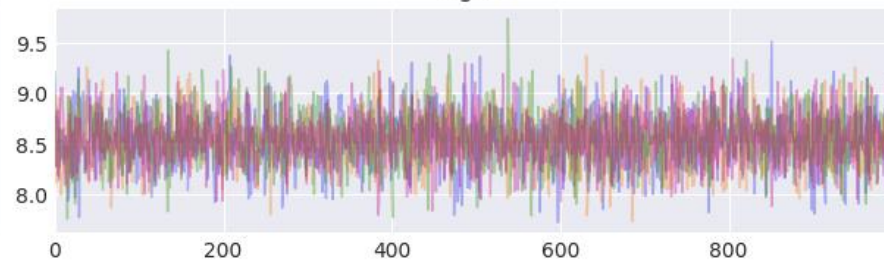
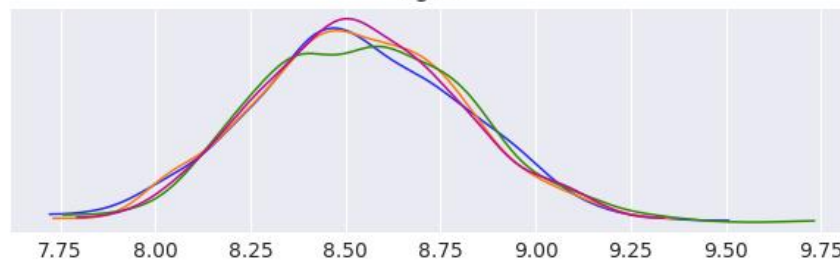
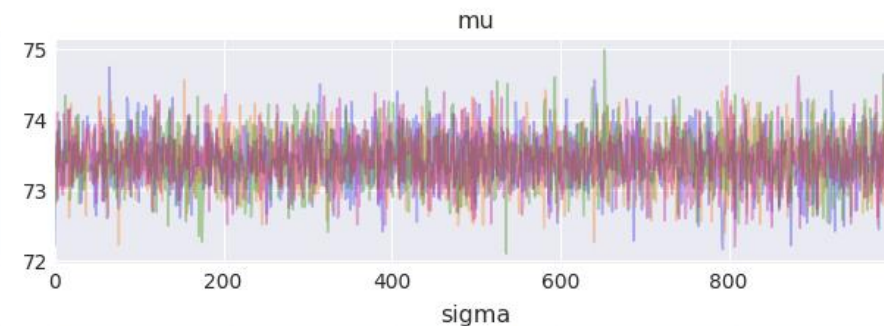
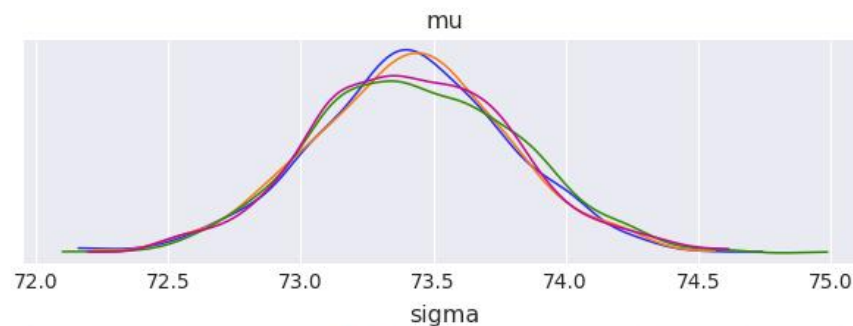
```
with normal_before_model:  
    idata_normal_before_prior = pm.sample_prior_predictive(  
        samples=1000, random_seed=random_seed)  
  
az.plot_ppc(idata_normal_before_prior, group="prior",  
            observed=True, num_pp_samples=100, random_seed=random_seed)
```



Normal Distribution Bayesian Model

- Sampling diagnostic
 - Summary diagnostics
 - Trace plots

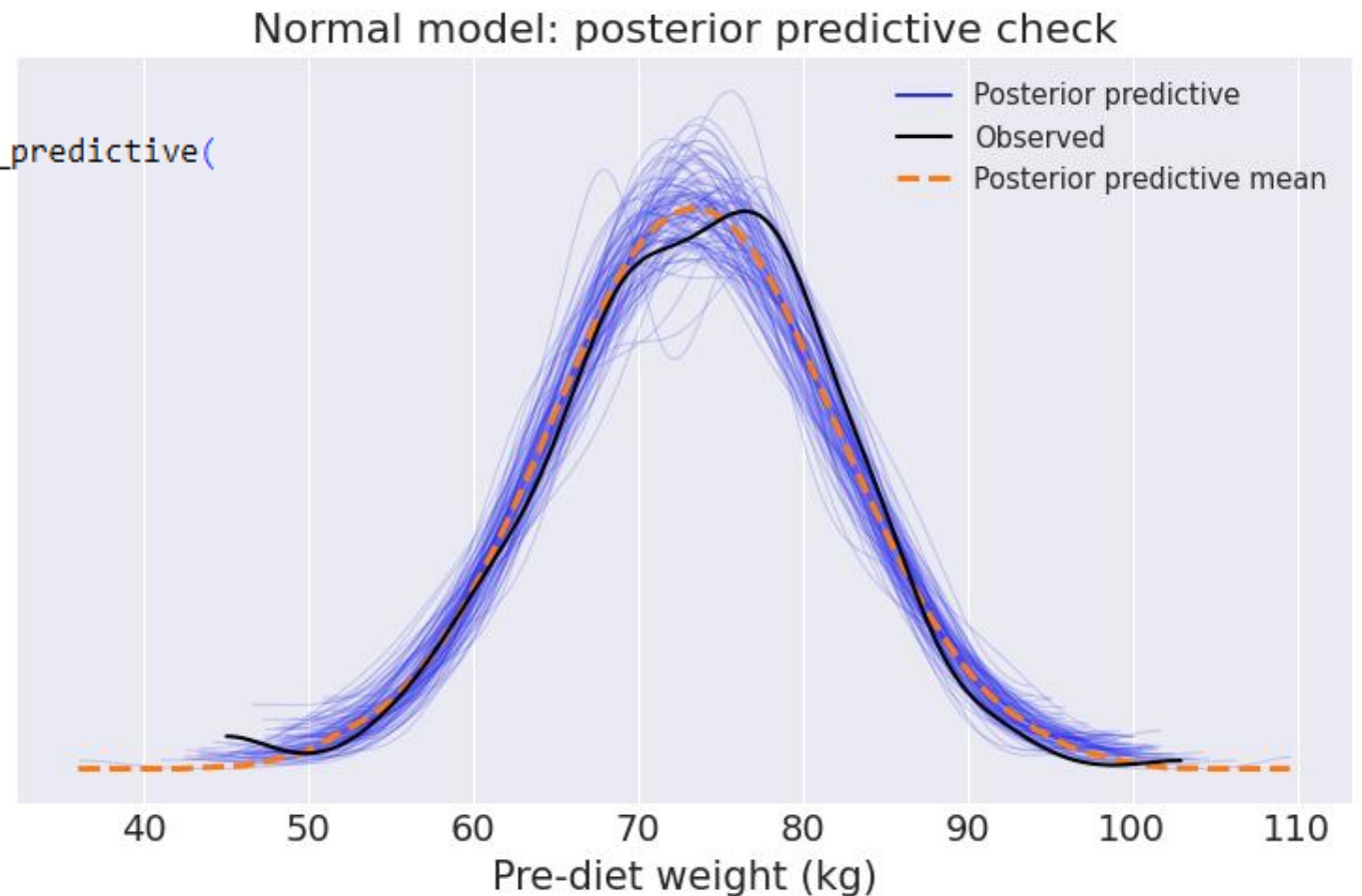
	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu	0.006	0.006	4275.0	3084.0	1.0
sigma	0.004	0.004	4194.0	2896.0	1.0



Normal Distribution Bayesian Model

- Posterior predictive check

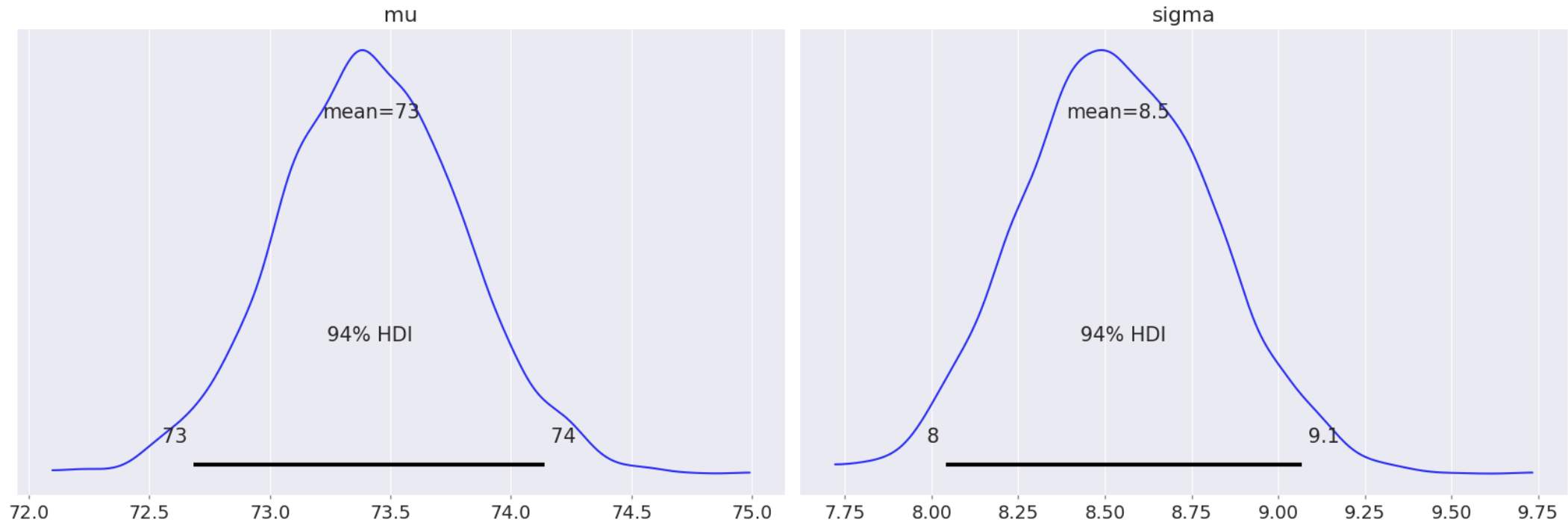
```
with normal_before_model:  
    idata_normal_before = pm.sample_posterior_predictive(  
        idata_normal_before,  
        var_names=["y_obs"],  
        random_seed=random_seed,  
        extend_inferencedata=True  
    )  
  
az.plot_ppc(  
    idata_normal_before,  
    group="posterior",  
    observed=True,  
    num_pp_samples=100,  
    random_seed=random_seed  
)
```



Normal Distribution Bayesian Model

- Posterior

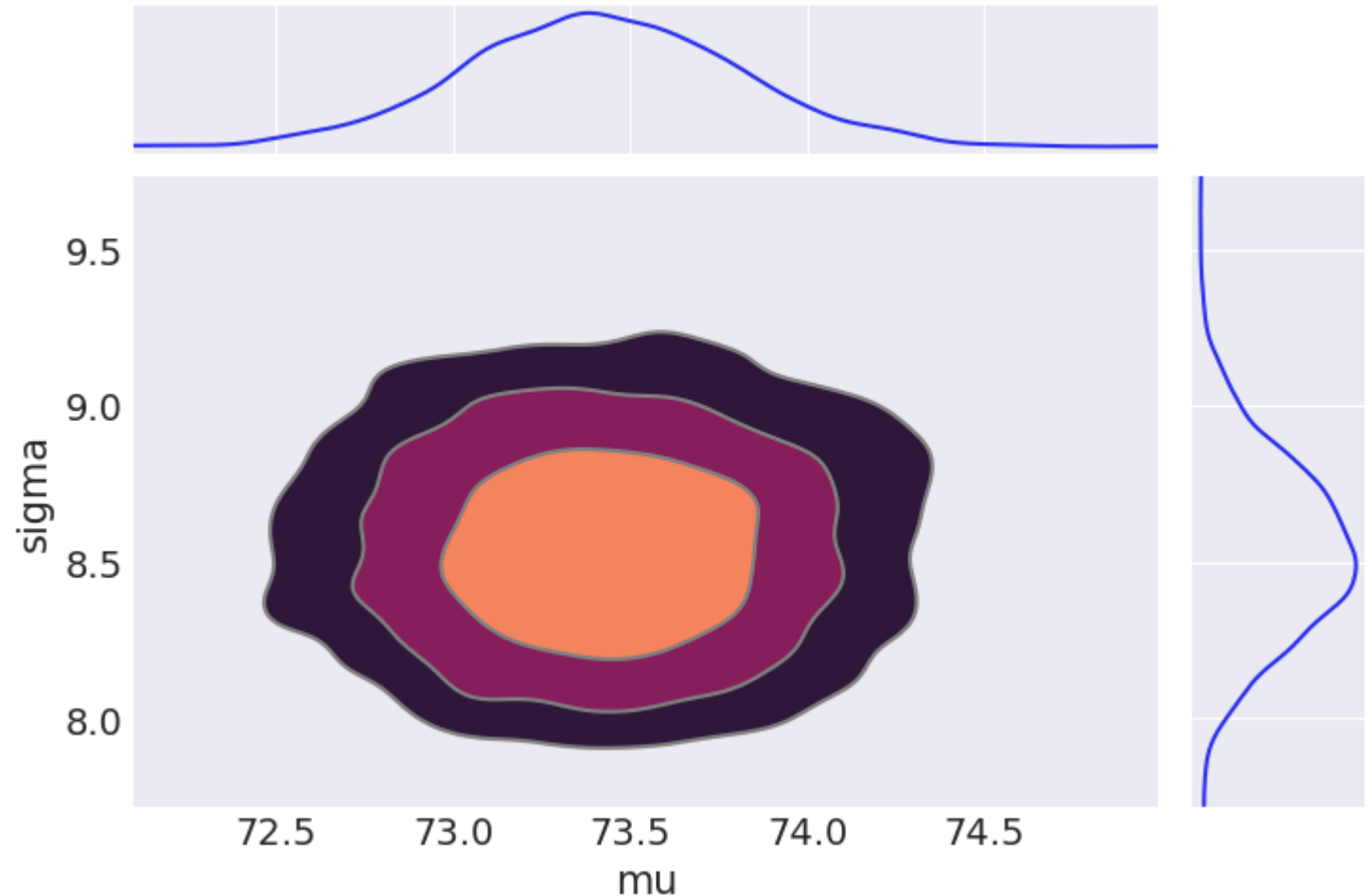
	mean	sd	hdi_3%	hdi_97%
mu	73.42	0.38	72.68	74.14
sigma	8.54	0.28	8.04	9.07



Normal Distribution Bayesian Model

- **Posterior**
 - Pair plot

```
az.plot_pair(  
    idata_normal_before,  
    var_names=["mu", "sigma"],  
    kind="kde",  
    marginals=True  
)
```



Bayesian p-value

- Based on posterior predictive samples
- For each posterior predictive dataset, we calculate the same statistic as in the observed data.
- Then we compare the observed statistic to the posterior predictive sampling distribution.

Bayesian p-value

- Similarity to frequentist p-values:
 - Both ask whether the observed data are extreme relative to a reference distribution.
- Difference from frequentist p-values:
 - Frequentist p-values integrate the null sampling distribution.
 - Bayesian p-values integrate, or sum over, the posterior predictive sampling distribution.
 - Bayesian p-values are used as a measure of fit, not as a significance decision.

Normal Distribution Bayesian Model

- **How to interpret Bayesian p-values?**
- Values close to 0 or 1 may indicate that the model has difficulty reproducing that aspect of the observed data.
- Values closer to the middle suggest that the observed statistic is not unusual under the fitted model.

Normal Distribution Bayesian Model

- **Bayesian p-value – sample mean**

```
# Bayesian p value for the sample mean
normal_before_ppc = posterior_predictive_matrix(
    idata=idata_normal_before,
    var_name="y_obs",
    obs_dim="obs_before")

observed_mean_before = before.mean()
replicated_means_before = normal_before_ppc.mean(axis=1)

bayesian_p_value_mean = np.mean(replicated_means_before <= observed_mean_before)

print(f"Observed mean: {observed_mean_before:.2f} kg")
print(f"Bayesian p-value for the mean: {bayesian_p_value_mean:.3f}")

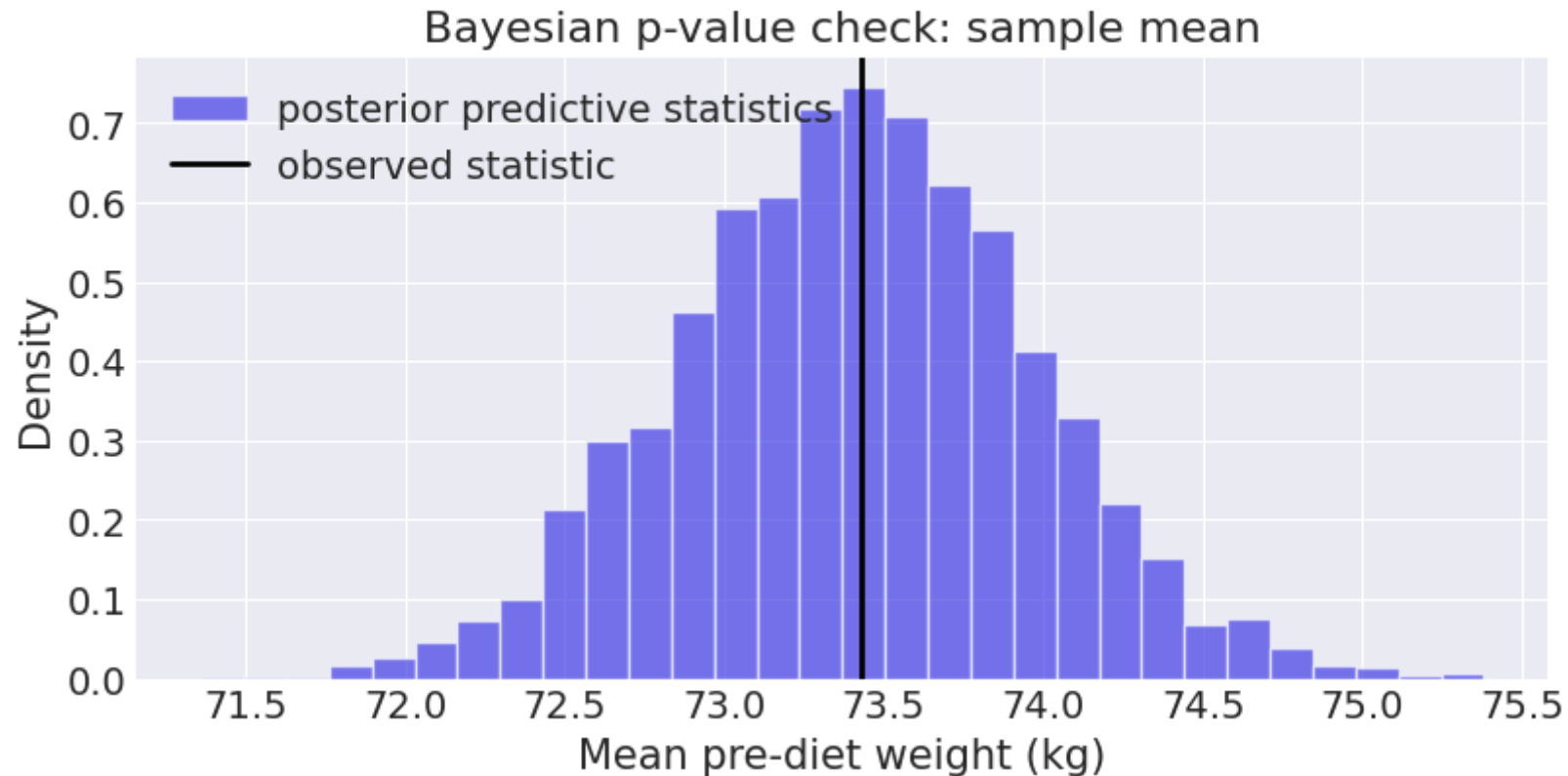
plot_observed_vs_replicated_statistic(
    replicated_statistics=replicated_means_before,
    observed_statistic=observed_mean_before,
    statistic_name="Mean pre-diet weight (kg)",
    title="Bayesian p-value check: sample mean")
```

Normal Distribution Bayesian Model

- **Bayesian p-value – sample mean**

Observed mean: 73.43 kg

Bayesian p-value for the mean: 0.506



Normal Distribution Bayesian Model

- **Bayesian p-value -sample SD**

```
# Bayesian p value for the sample std
observed_sd_before = before.std(ddof=1)
replicated_sds_before = normal_before_ppc.std(axis=1, ddof=1)

bayesian_p_value_sd = np.mean(replicated_sds_before <= observed_sd_before)

print(f"Observed SD: {observed_sd_before:.2f} kg")
print(f"Bayesian p-value for the SD: {bayesian_p_value_sd:.3f}")

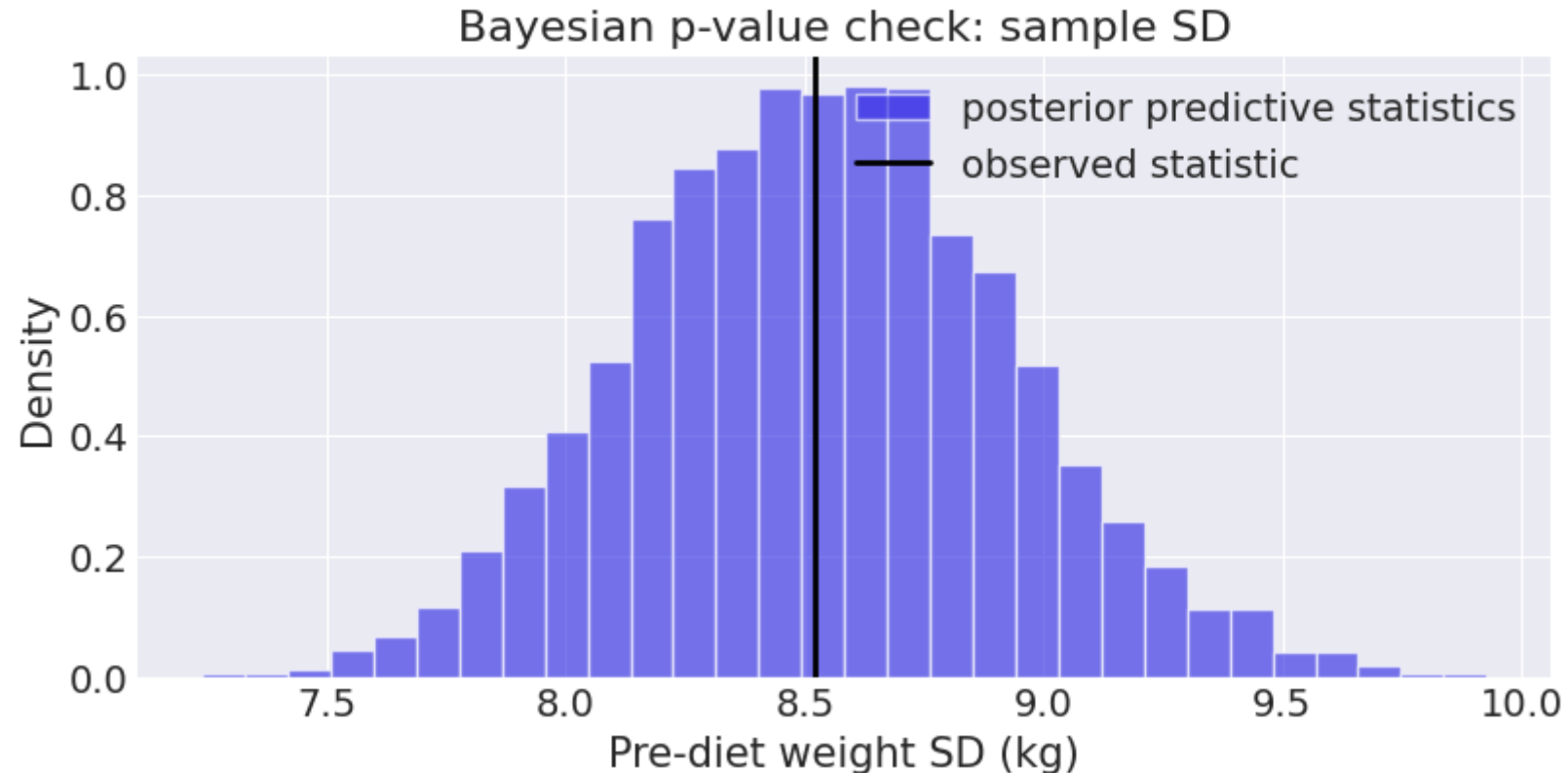
plot_observed_vs_replicated_statistic(
    replicated_statistics=replicated_sds_before,
    observed_statistic=observed_sd_before,
    statistic_name="Pre-diet weight SD (kg)",
    title="Bayesian p-value check: sample SD")
```


Normal Distribution Bayesian Model

- **Bayesian p-value - sample SD**

Observed SD: 8.52 kg

Bayesian p-value for the SD: 0.489



Paired vs. Unpaired Data

Paired Data

- Two measurements belong to the same participant:

$$y_{i,before}, y_{i,after}$$

- We can convert to change scores:

$$change_i = y_{i,before} - y_{i,after}$$

- And model the changes:

$$change_i \sim \text{Normal}(\mu_{\text{change}}, \sigma_{\text{change}})$$

Unpaired Data

- **Unpaired data**
- Different participants belong to different groups:

$$y_i \sim \text{Normal}(\mu_{\text{group}[i]}, \sigma)$$

Bayesian Paired Model & Hypothesis testing

Bayesian Paired Model

- The food diet data are paired because each person was measured before and after six weeks.

$$change_i = after_i - before_i$$

- The paired question is:
 - Is the average within-person weight change different from zero?
- Null hypothesis

$$H_0: \Delta_\mu = 0$$

- Alternative hypothesis

$$H_1: \Delta_\mu \neq 0$$

Effect size - Cohen's d

Raw effect size

$$\Delta_{\mu} = \mu_A - \mu_B$$

- This is in original units.

Cohen's d

$$\text{Cohen's } d = \frac{\mu_A - \mu_B}{\sigma}$$

- Standardized effect size.

Cohen's d	
0.2	Small
0.5	Medium
0.8	Large

Reminder- ROPE and HDI

HDI

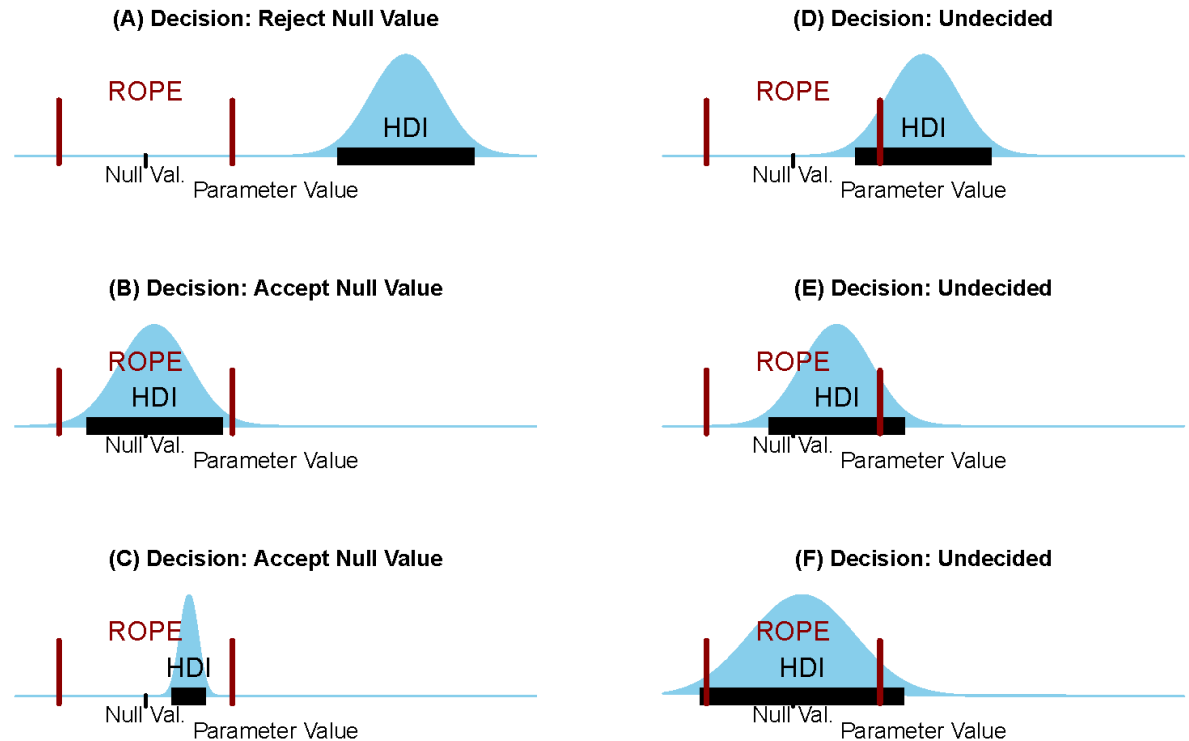
- The interval containing the most credible parameter values.

ROPE

- A region of practical equivalence:
 $[-\epsilon, \epsilon]$

Reminder- ROPE and HDI Test

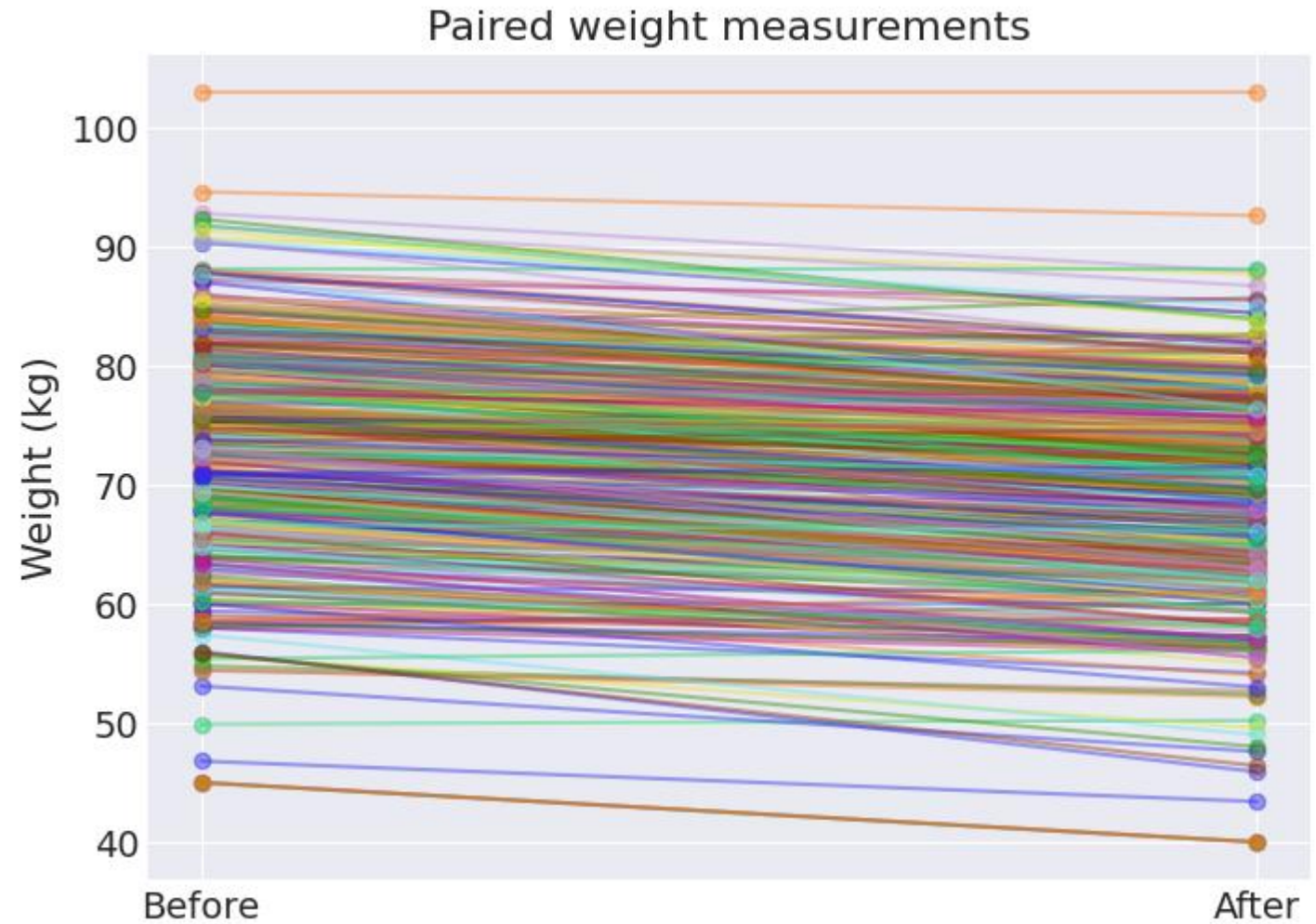
- The ROPE is not a mathematical default. It is a domain-based choice.
- Decision logic:
 - HDI completely **outside** ROPE → reject the practical null.
 - HDI completely **inside** ROPE → accept the practical null.
 - HDI partly **overlaps** ROPE → no clear decision.



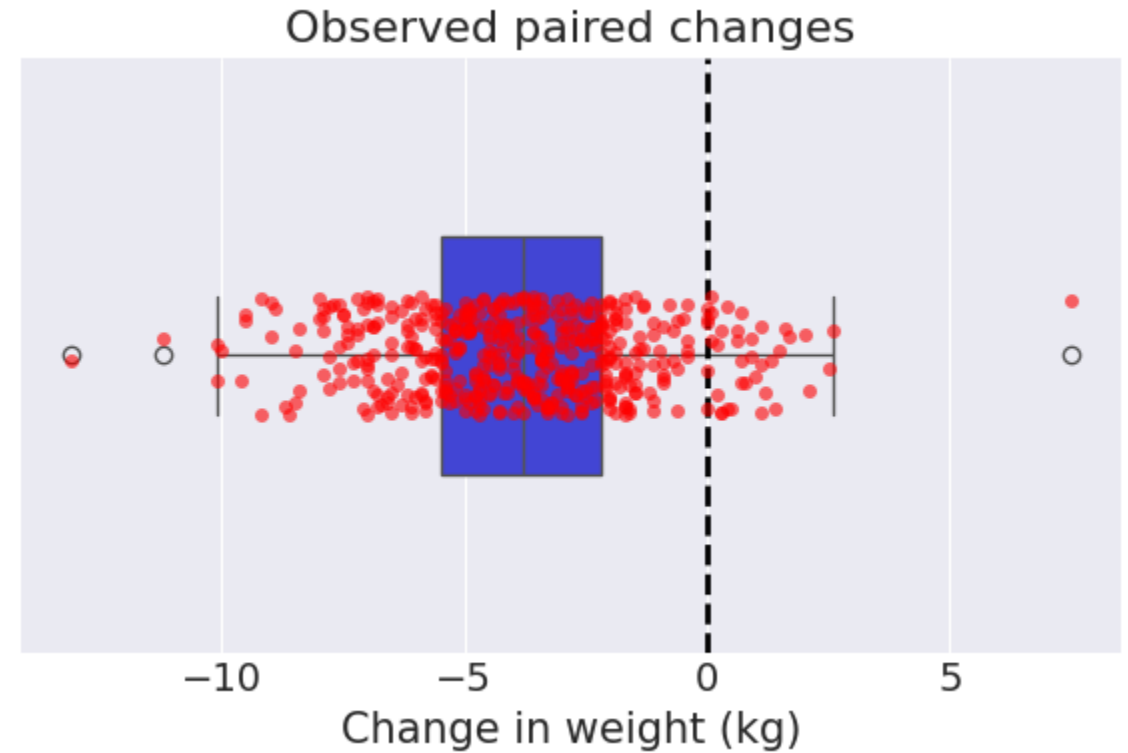
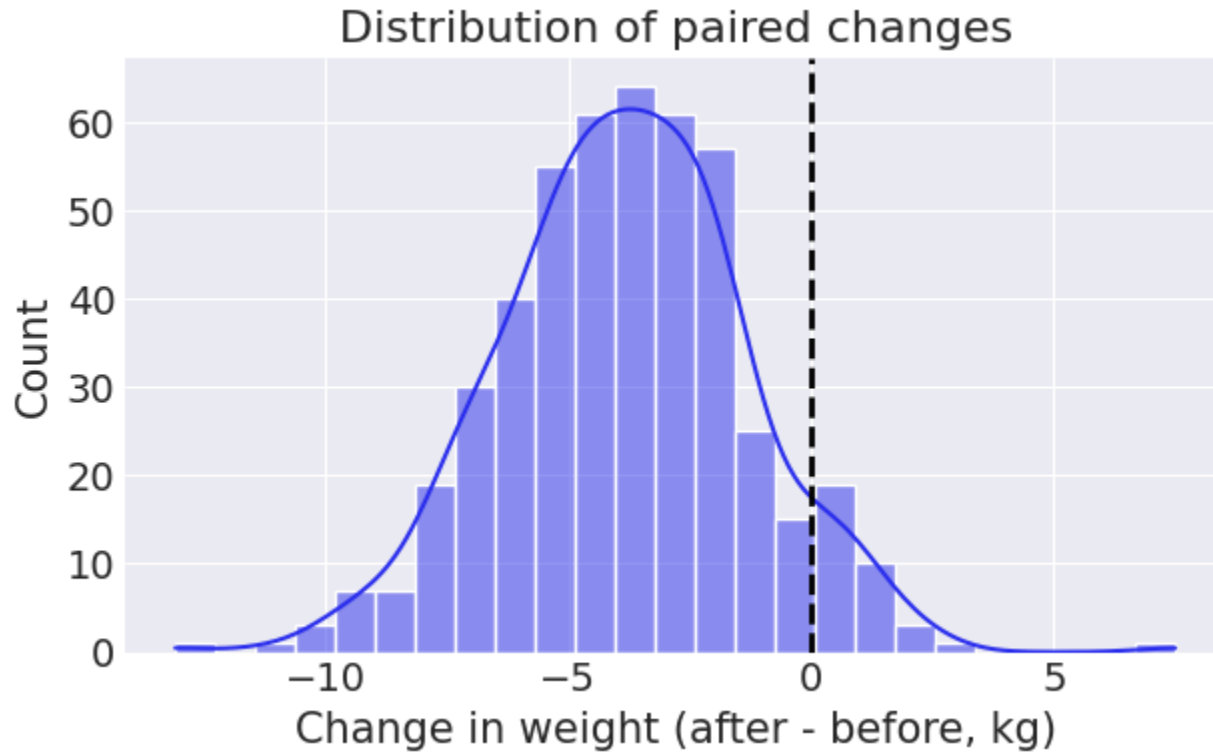
Bayesian Paired Model

- Visualize the paired data:

	Measure	Mean (kg)	SD (kg)
0	Before diet	73.43	8.52
1	After 6 weeks	69.58	8.62
2	Change: after - before	-3.85	2.53



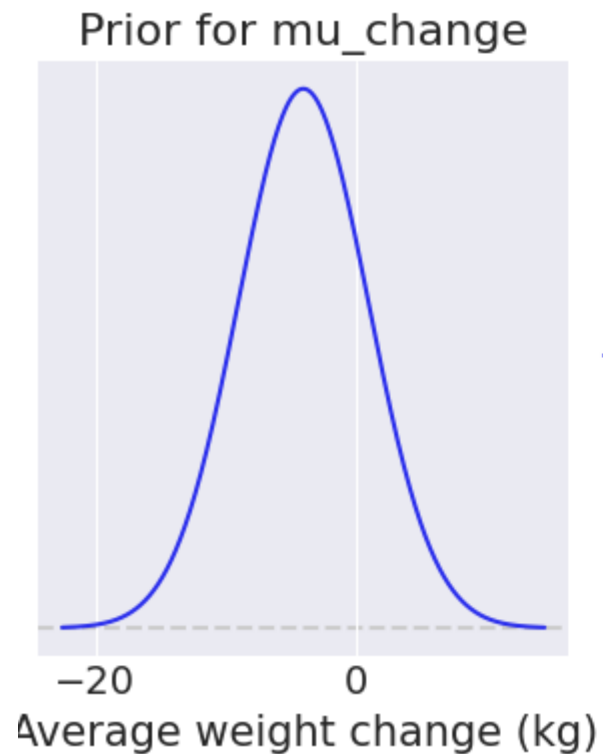
Bayesian Paired Model



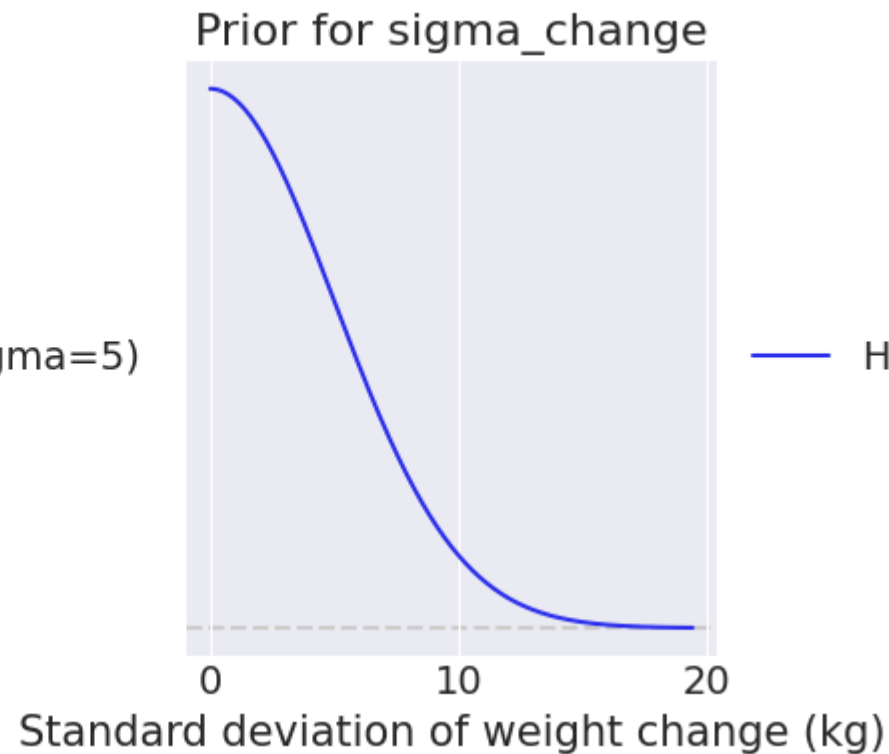
Bayesian Paired Model

- Choosing priors for the change

```
change_prior_mean = -4  
change_prior_sigma = 5  
change_sigma_prior = 5
```



— Normal($\mu=-4$, $\sigma=5$)



— HalfNormal($\sigma=5$)

Bayesian Paired Model

- Model

```
coords_food_change = {"obs": np.arange(len(change))}

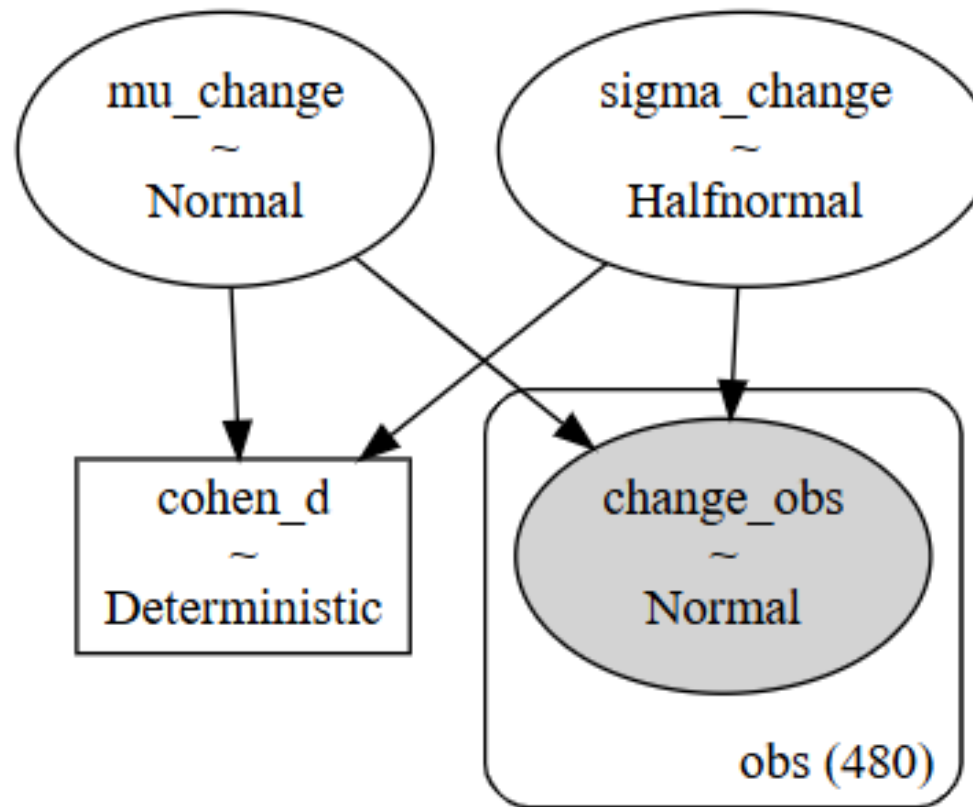
with pm.Model(coords=coords_food_change) as paired_normal_model:
    # Priors
    mu_change = pm.Normal("mu_change", mu=change_prior_mean, sigma=change_prior_sigma)
    sigma_change = pm.HalfNormal("sigma_change", sigma=change_sigma_prior)
    # effect size (Deterministic)
    cohen_d = pm.Deterministic("cohen_d", mu_change / sigma_change)
    # Likelihood
    change_obs = pm.Normal("change_obs", mu=mu_change, sigma=sigma_change, observed=change, dims="obs")

pm.model_to_graphviz(paired_normal_model)
```

Bayesian Paired Model

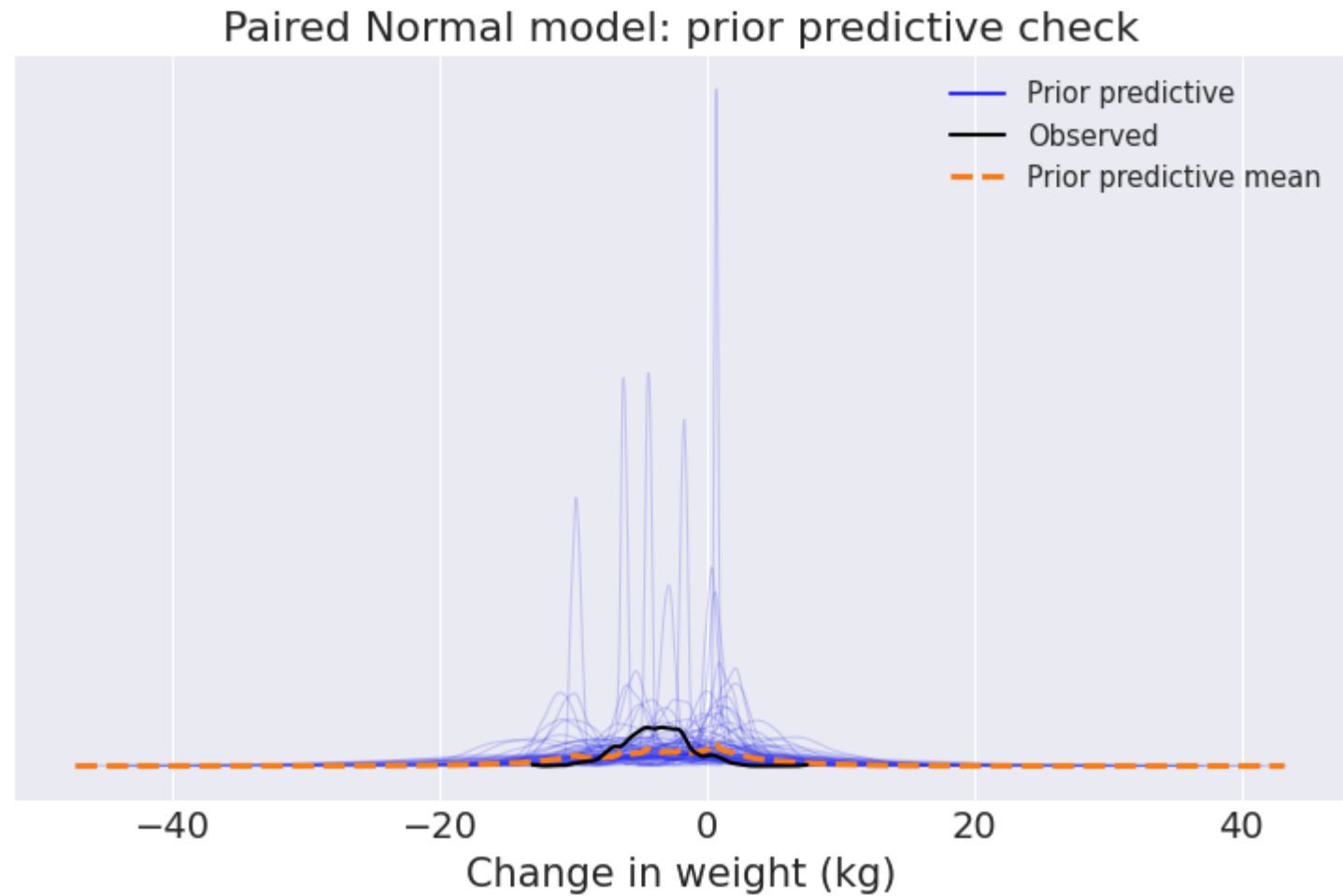
- Model

```
pm.model_to_graphviz(paired_normal_model)
```



Bayesian Paired Model

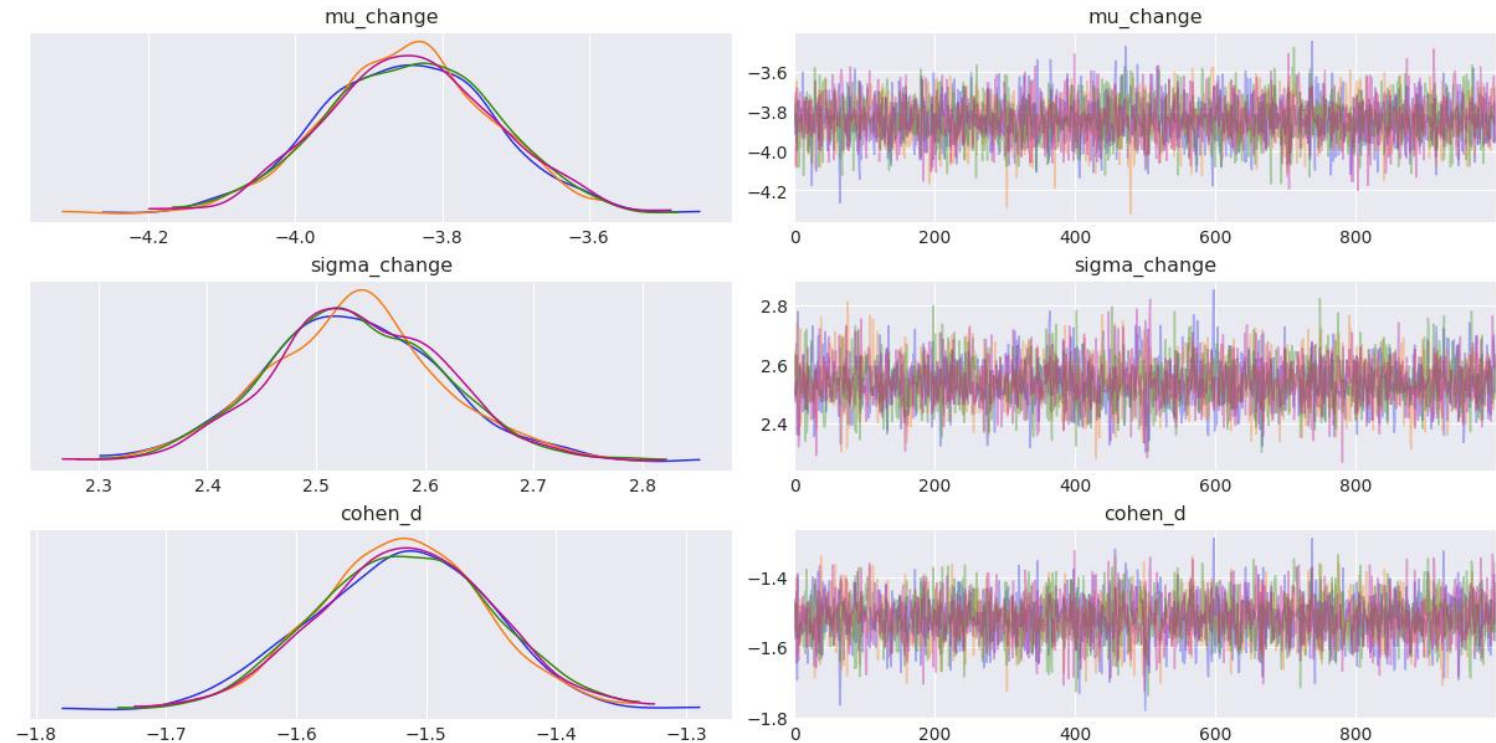
- Prior predictive



Bayesian Paired Model

- Diagnostics

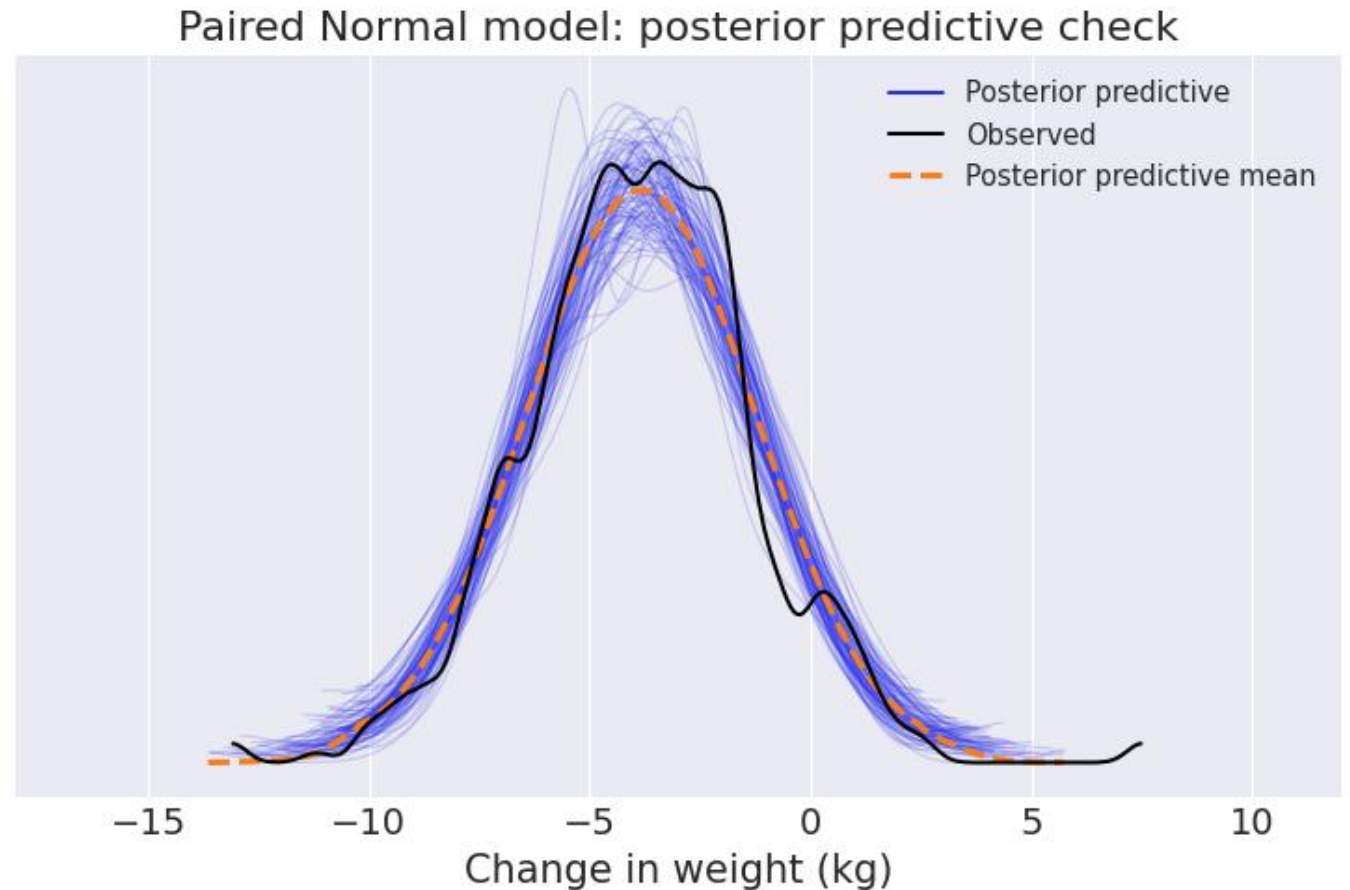
	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_change	0.002	0.002	3905.0	2987.0	1.0
sigma_change	0.001	0.001	4262.0	2942.0	1.0
cohen_d	0.001	0.001	4119.0	3154.0	1.0



Bayesian Paired Model

- Posterior Predictive

```
with paired_normal_model:  
    idata_paired_normal = pm.sample_posterior_predictive(  
        idata_paired_normal,  
        var_names=["change_obs"],  
        random_seed=random_seed,  
        extend_inferencedata=True  
    )  
  
az.plot_ppc(  
    idata_paired_normal,  
    group="posterior",  
    observed=True,  
    num_pp_samples=100,  
    random_seed=random_seed  
)
```



Bayesian Paired Model

- Posterior

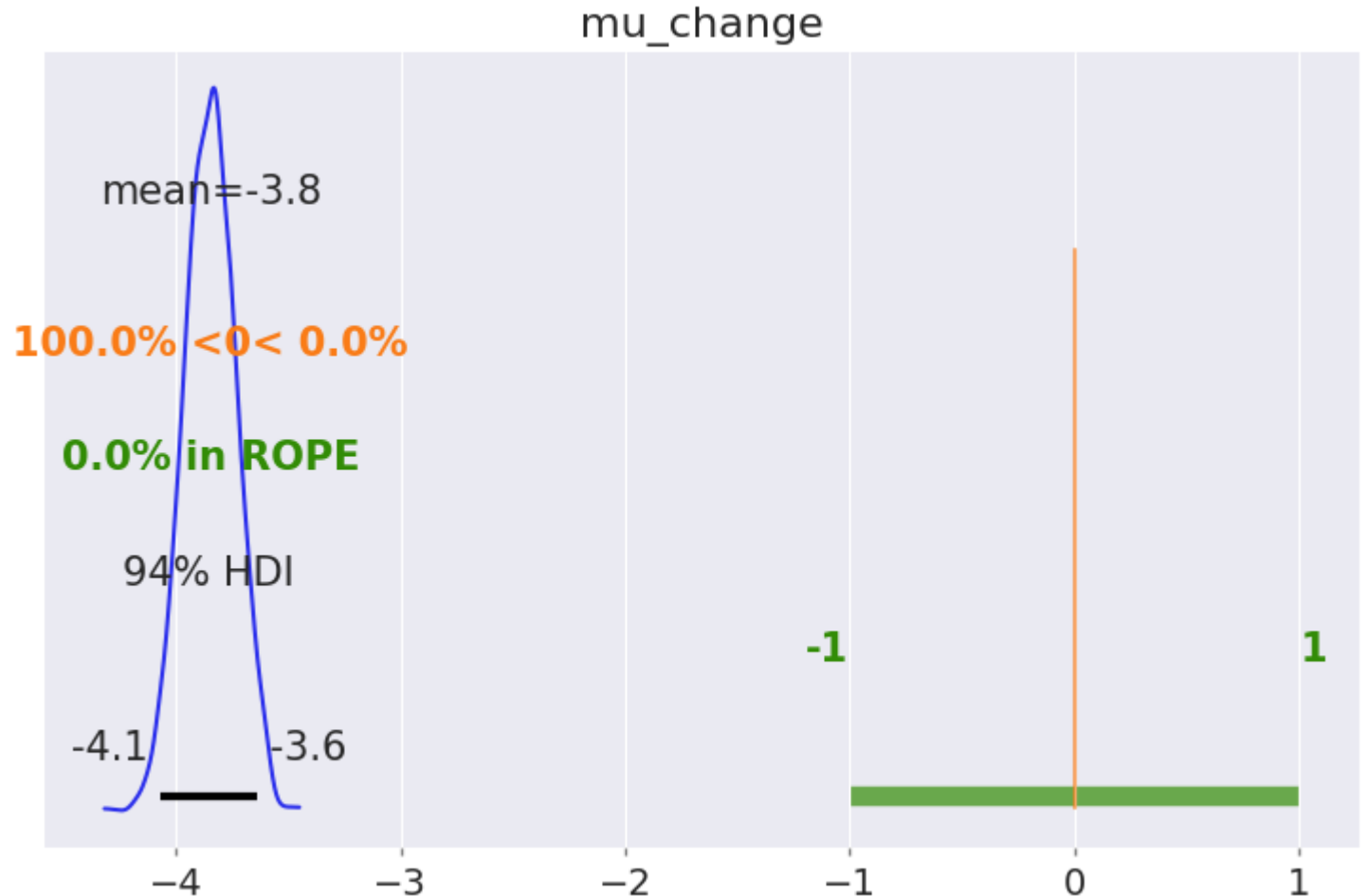
	mean	sd	hdi_3%	hdi_97%
mu_change	-3.85	0.11	-4.06	-3.64
sigma_change	2.54	0.08	2.38	2.69
cohen_d	-1.52	0.07	-1.65	-1.40

Bayesian Paired Model

- HDI\ROPE test

```
rope_change = (-1, 1)
```

```
az.plot_posterior(  
    idata_paired_normal,  
    var_names=["mu_change"],  
    ref_val=0, rope=rope_change)
```



Bayesian Paired Model

- HDI\ROPE test

	quantity	posterior mean	posterior median	94% HDI low	94% HDI high	Pr(value > 0)	Pr(value < 0)	Pr(value in ROPE)
0	Mean change after - before	-3.847	-3.845	-4.064	-3.637	0.000	1.000	0.000
1	Cohen's d for paired change	-1.518	-1.517	-1.648	-1.398	0.000	1.000	nan

Bayesian Paired Model

- **Reporting**
- The observed mean change after six weeks was -3.85 kg.
- The Bayesian paired Normal model estimated the posterior mean change as -3.85 kg, with a 94% HDI from -4.06 to -3.64 kg.
- The posterior probability that the change is positive was 0.000.

Frequentist t-tests

- Frequentist t-tests are used to test hypotheses about means of continuous variables.
- They compare an observed mean difference to the variation expected under the null hypothesis.

Frequentist t-tests

- A t-test assumes the null hypothesis is true.
- Under H_0 , the test statistic follows a t-distribution:

$$t = \frac{\textit{observed estimate} - \textit{null value}}{\textit{standard error}}$$

- The t-distribution is the null sampling distribution of the t-statistic.
- We ask:
 - If H_0 were true, how extreme would our observed t-statistic be?
- A very extreme t-statistic gives a small p-value.

Frequentist t-tests

- **Why do we use the t-distribution?**
- The z-test uses a Normal reference distribution.
- The t-test uses a t-distribution because the population standard deviation is unknown and estimated from the sample.

$$t \sim t_{df}$$

- The t-distribution:
 - Is centered at 0 under H_0
 - Has heavier tails than the Normal distribution
 - Depends on degrees of freedom
 - Becomes closer to Normal as df increases

Frequentist t-tests

- **One-sample t-test**

- Tests whether one mean differs from a reference value.

$$H_0: \mu = \mu_0$$

- **Paired t-test**

- Tests whether the mean within-subject change differs from zero.

$$H_0: \mu_{\text{change}} = 0$$

- **Independent two-sample t-test**

- Tests whether two independent group means differ.

$$H_0: \mu_A - \mu_B = 0$$

- **Welch t-test**

- Same mean comparison as the independent t-test, but does not assume equal variances.

- **Mann–Whitney / Wilcoxon**

- Rank-based alternatives

t-Test Overview

Test Type	Common H_0	Description	Example
One-sample t-test	$H_0: \mu = \mu_0$	One continuous variable compared to a reference value	Is average reaction time different from 300 ms?
Paired two-sample t-test	$H_0: \mu_{\text{change}} = 0$	Same subjects measured twice, or matched pairs	Pre/post-treatment, left vs. right
Independent two-sample t-test	$H_0: \mu_A - \mu_B = 0$	Two separate groups	Control vs. experimental group
Welch t-test	$H_0: \mu_A - \mu_B = 0$	Two independent groups, unequal variances allowed	Same comparison, but group spreads may differ
Mann–Whitney	Distributions differ between groups	Two independent groups	Rank-based independent-group alternative
Wilcoxon	Median paired change =0	Paired / repeated measurements	Rank-based paired alternative

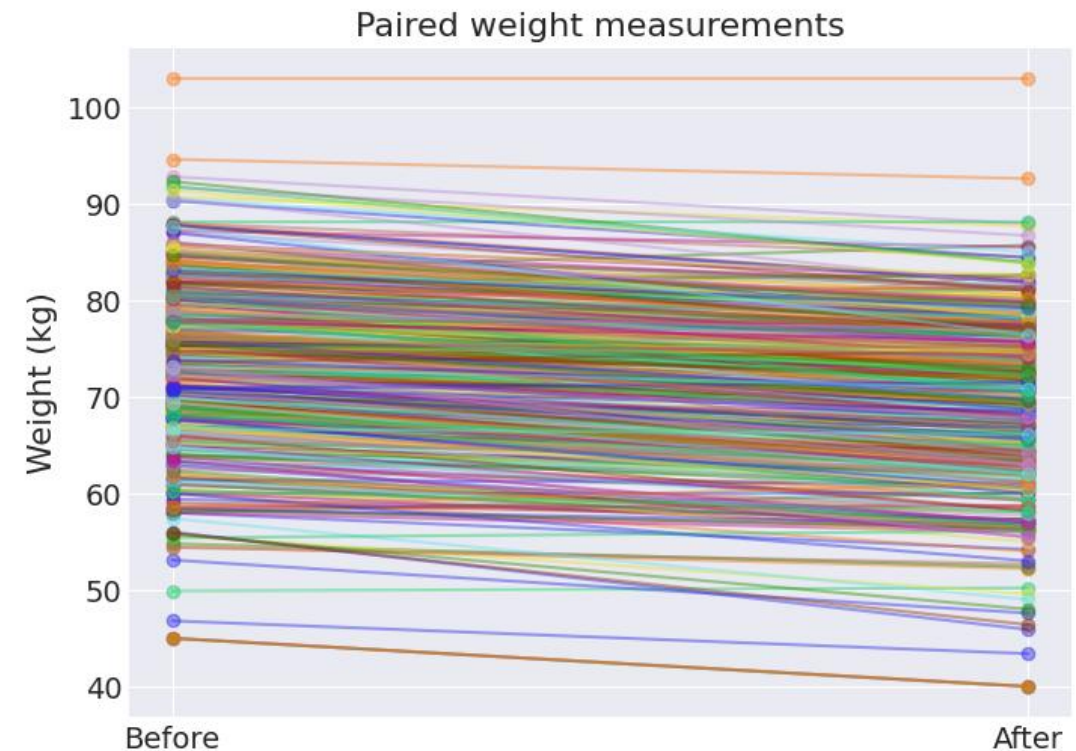
t-Test

Assumptions:

- The outcome variable is **continuous** (not categorical)
- Data (or paired differences) are **approximately normal**
- **Observations are independent** (for independent t-test)
- **Equal variances between groups** (or use Welch's correction)

Paired t-Test

- foodDiet_extended.csv dataset:
 - Same participants measured before and after diet
 - Tests whether mean weight changed significantly after the diet



Paired t-Test

- The paired t-test asks whether the average within-subject change is different from zero.
- For each subject:

$$\text{change}_i = \text{after}_i - \text{before}_i$$

- Null hypothesis:

$$H_0: \mu_{\text{change}} = 0$$

- In paired t-test :

$$t \sim t_{df}, \quad df = n - 1$$

- Test statistic:

$$t = \frac{\bar{x}_{\text{change}} - 0}{\frac{s_{\text{change}}}{\sqrt{n}}}$$

where:

$\bar{x}_{\text{change}}$ is the average observed change

s_{change} is the standard deviation of the changes

n is the number of paired observations

- Decision:

$$p < \alpha \Rightarrow \text{Reject } H_0$$

Paired t-Test

```
paired_t_result = stats.ttest_rel(after, before)
wilcoxon_result = stats.wilcoxon(change, zero_method="wilcox")
```

```
mean_change = change.mean()
sd_change = change.std(ddof=1)
n_change = len(change)
se_change = sd_change / np.sqrt(n_change)
```

```
t_critical_change = stats.t.ppf(1 - SIGNIFICANCE_ALPHA / 2, df=n_change - 1)
ci_change_low = mean_change - t_critical_change * se_change
ci_change_high = mean_change + t_critical_change * se_change
```

```
cohens_d_paired = mean_change / sd_change
```

scipy.stats.

ttest_rel

```
ttest_rel(a, b, axis=0, nan_policy='propagate', alternative='two-sided', *,
keepdims=False) \[source\]
```

Calculate the t-test on TWO RELATED samples of scores, a and b.

This is a test for the null hypothesis that two related or repeated samples have identical average (expected) values.

Parameters:

a, b : *array_like*

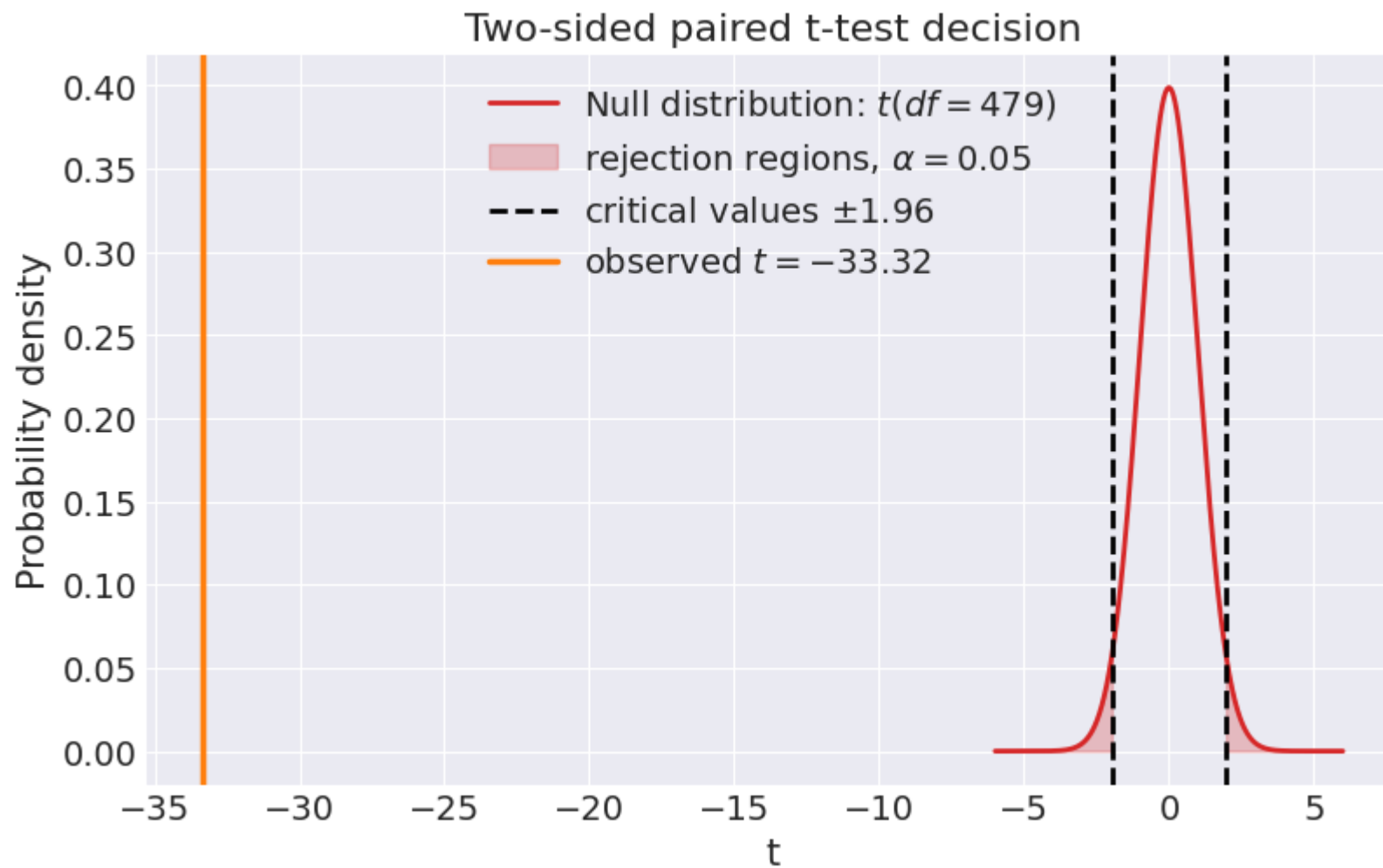
The arrays must have the same shape.

Paired t-Test

	test	statistic	p-value	mean change	CI low	CI high	Cohen's d
0	Paired t-test	-33.319	7.976e-127	-3.851	-4.078	-3.624	-1.521

- **t-statistic:** -33.319
 - **p-Value :** $7.976 \cdot 10^{-127}$ (< 0.05) $\Rightarrow$ low value, indicating strong statistical significance
 - **CI:** [-4.078, -3.624]
 - **Cohen's d:** -1.521 (large effect size)
- > Weight was significantly lower after the diet.

Paired t-Test



Paired t-Test

- **Reporting –**
- The observed mean change after six weeks was -3.85 kg (after - before), with a 95% confidence interval from -4.08 to -3.62 kg.
- The frequentist paired t-test gave $t(479) = -33.32$ and $p = 7.976e-127$.
Therefore, using $\alpha = 0.05$, we reject H_0 .

Paired Wilcoxon-Test

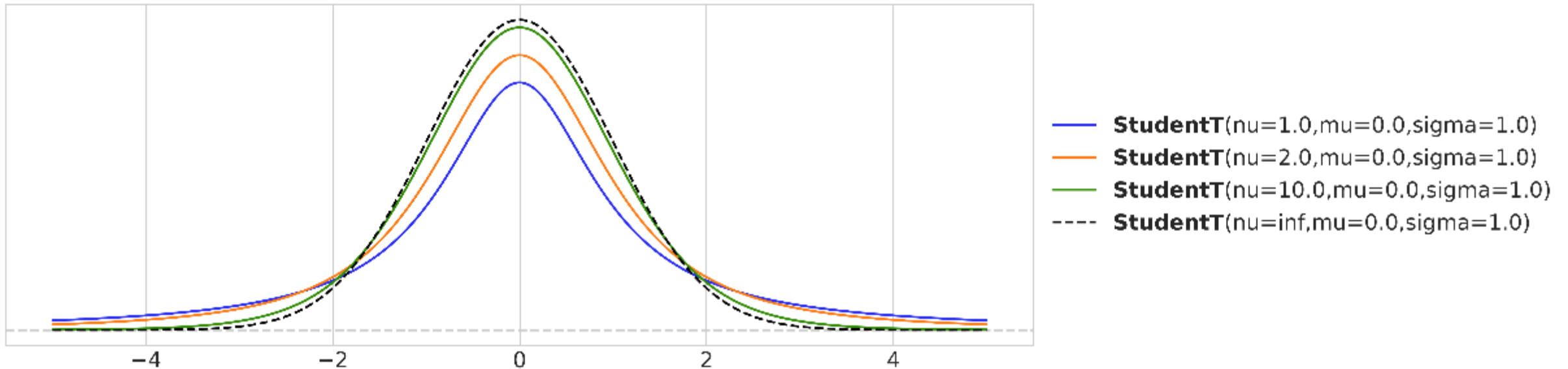
- Rank-based test
- It is used for paired data when we do not want to rely as strongly on the Normal model for the change scores. It tests whether the paired differences are systematically above or below zero.

```
wilcoxon_result = stats.wilcoxon(change, zero_method="wilcox")
```

	test	statistic	p-value	mean change	CI low	CI high	Cohen's d
0	Paired t-test	-33.319	7.976e-127	-3.851	-4.078	-3.624	-1.521
1	Wilcoxon signed-rank test	1713.000	6.581e-75	nan	nan	nan	nan

Student's t-distribution

- Similar to Normal distribution but with thicker tails
 - Has a parameter ν , which is the normality parameter (or degrees of freedom)
 - Converges to normal distribution as $\nu \rightarrow \infty$

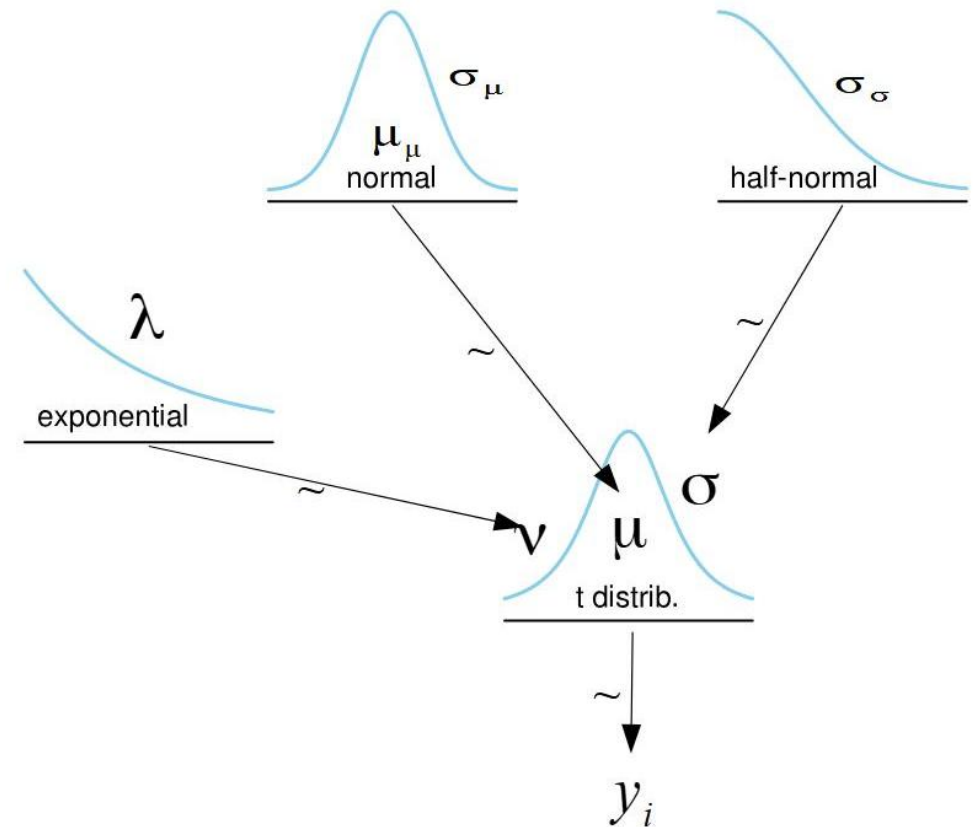


Student's t-distribution

- The thicker tails allows for capturing the outliers without necessitating an increase in the variance.
 - Higher ν can mean higher σ because the student's t distribution will be more like a normal distribution and capture the outlier information less well.
- Repeat our analysis with the student's t distribution

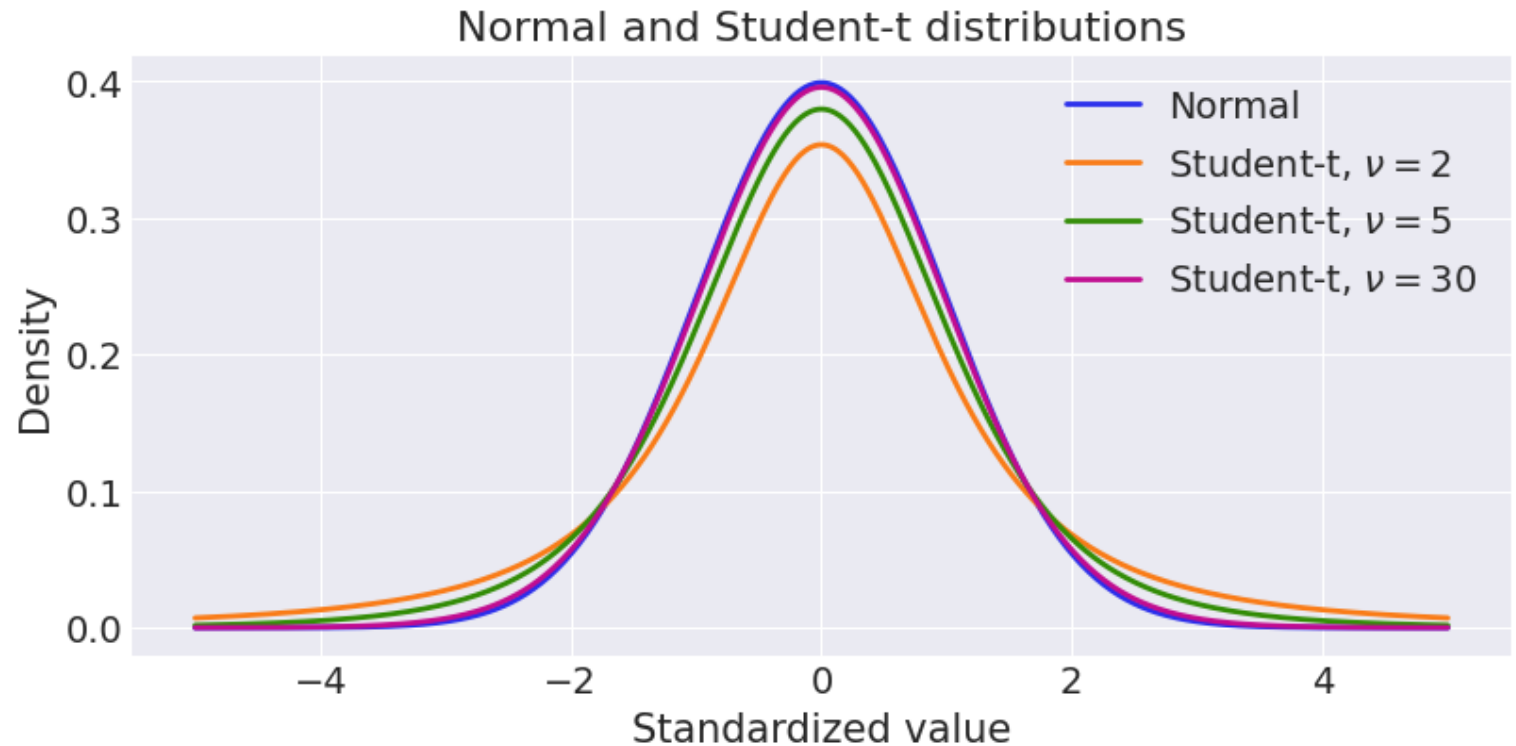
Student's t-distribution

- The thicker tails allows for capturing the outliers without necessitating an increase in the variance.
- The parameter ν controls tail thickness:
 - smaller ν : heavier tails
 - larger ν : closer to Normal



Student's t-distribution

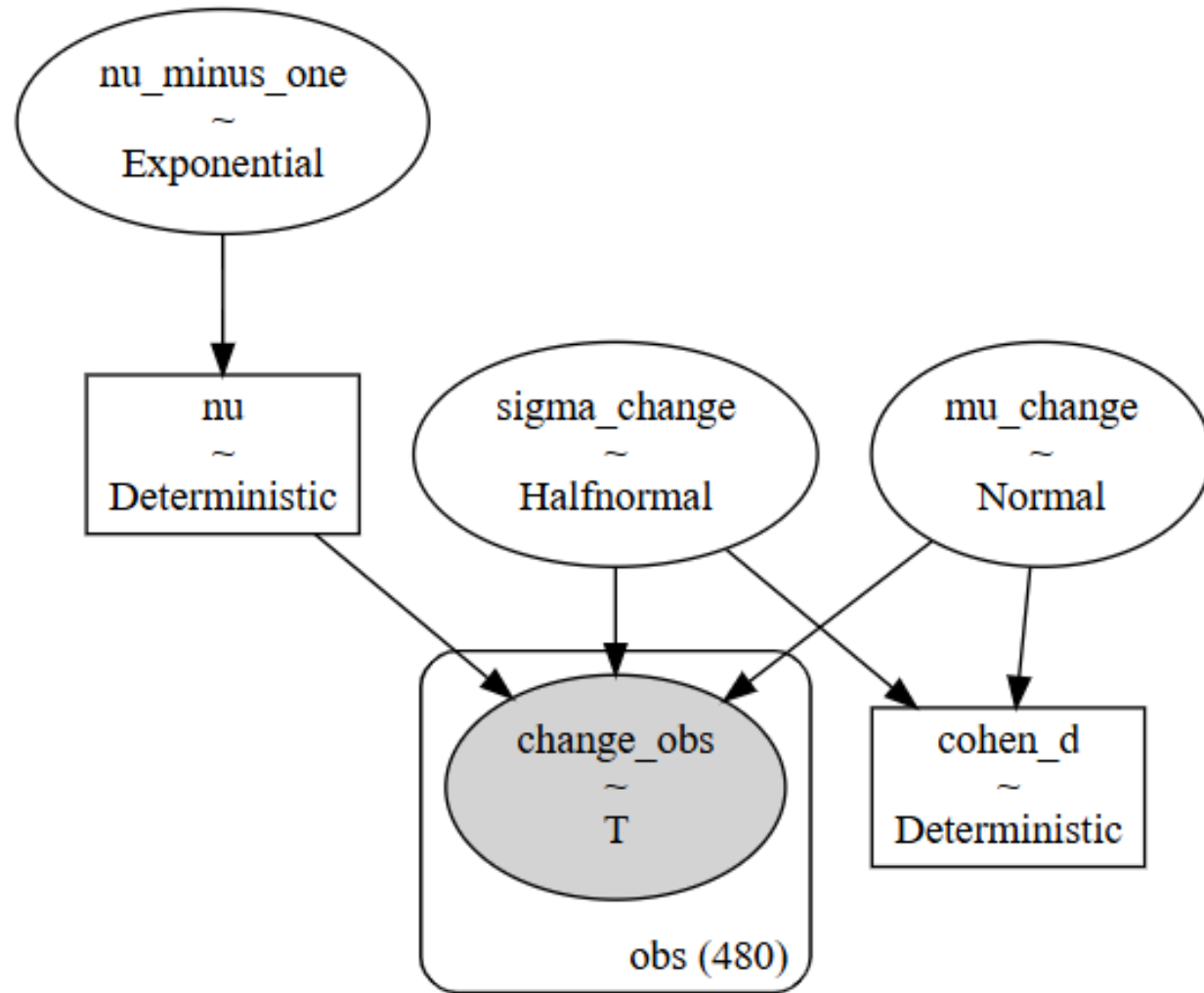
- The parameter ν controls tail thickness:
 - smaller ν : heavier tails
 - larger ν : closer to Normal



Robust Model

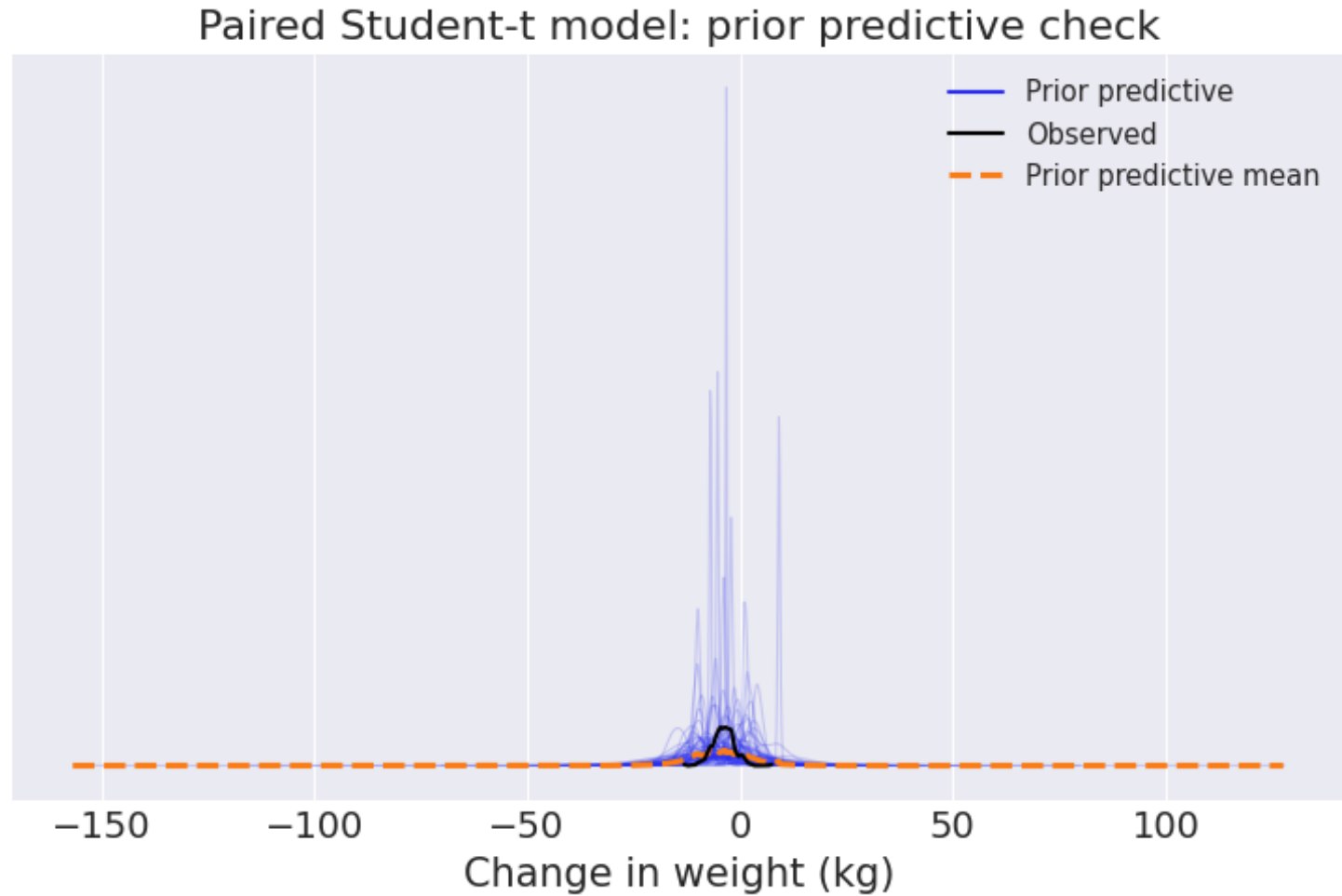
- Model

```
pm.model_to_graphviz(paired_student_t_model)
```



Robust Model

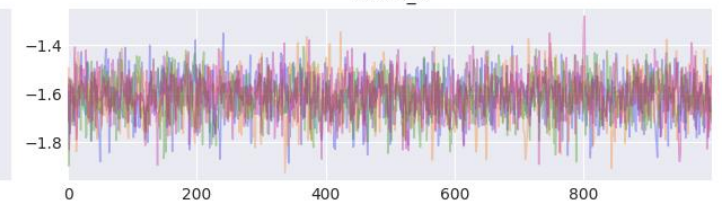
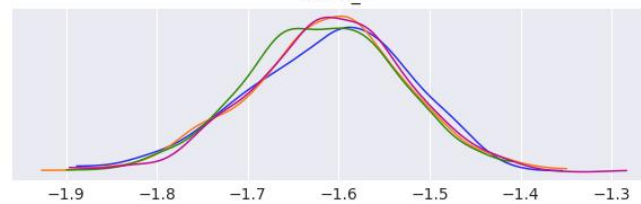
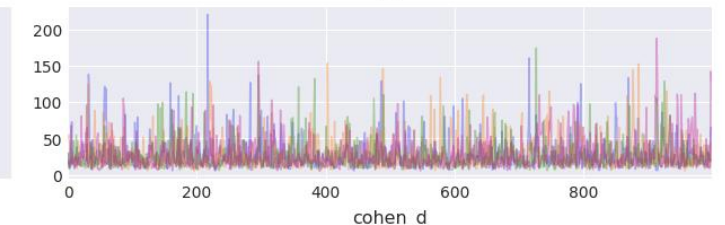
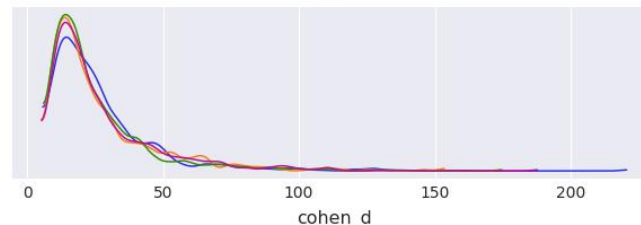
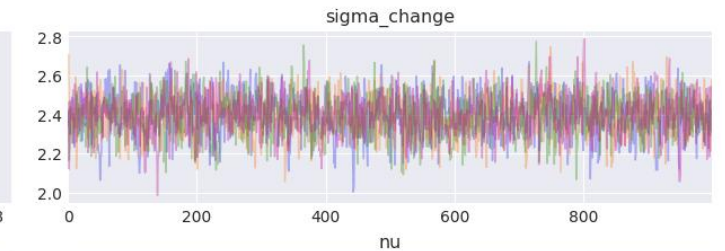
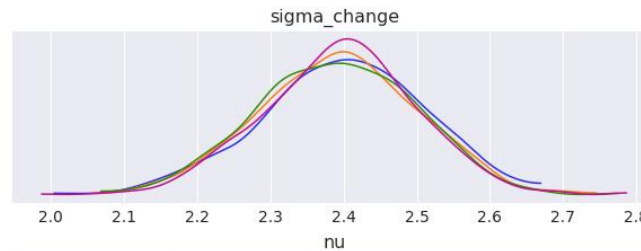
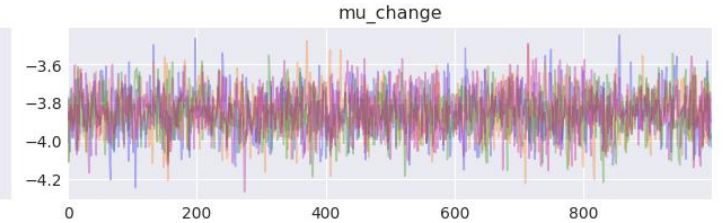
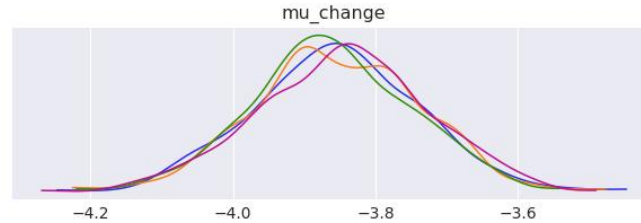
- Prior Predictive



Robust Model

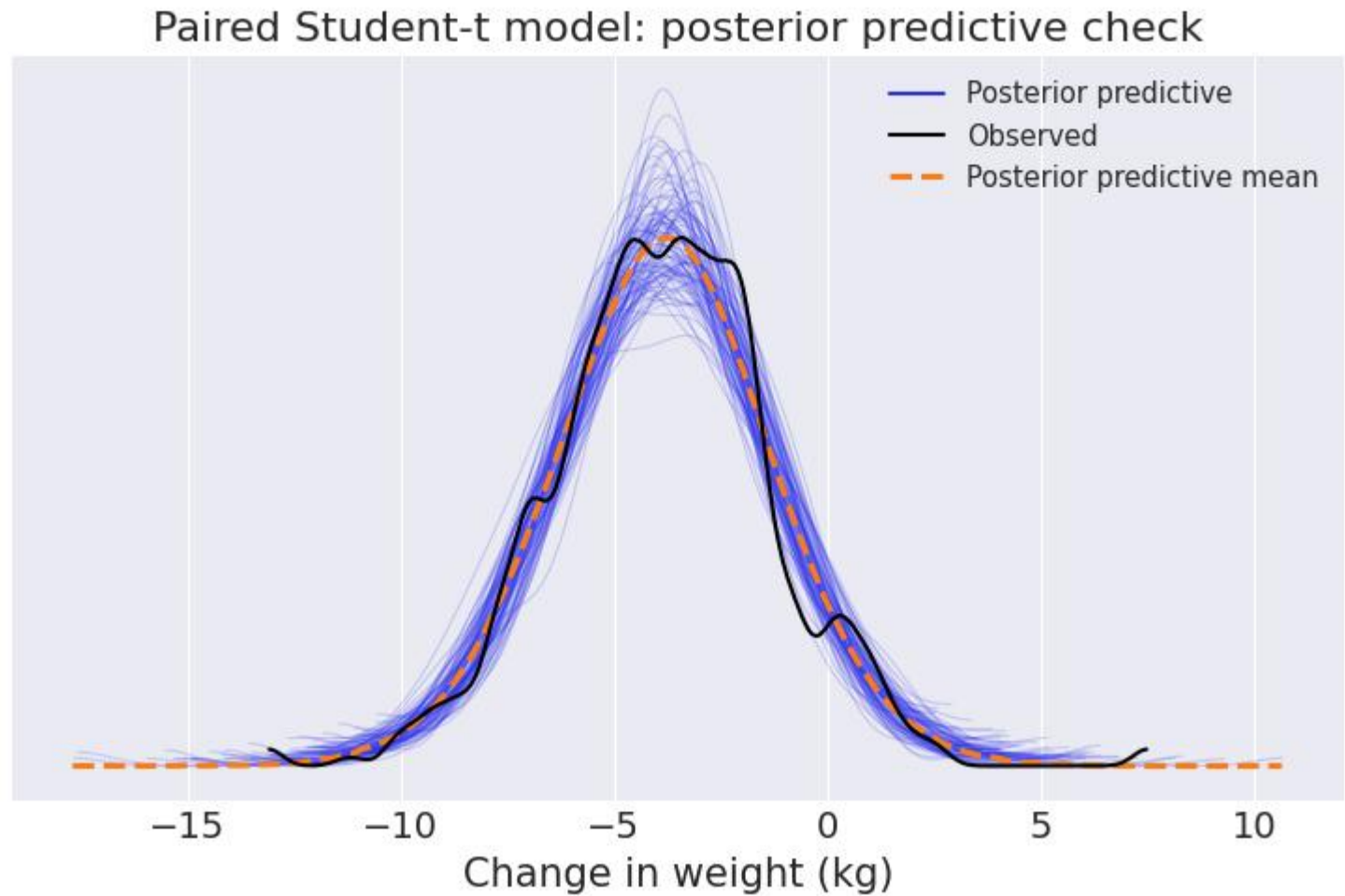
- Diagnostics

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_change	0.002	0.002	2539.0	2570.0	1.0
sigma_change	0.002	0.002	2226.0	2233.0	1.0
nu	0.445	0.705	2293.0	2376.0	1.0
cohen_d	0.002	0.001	2456.0	2355.0	1.0



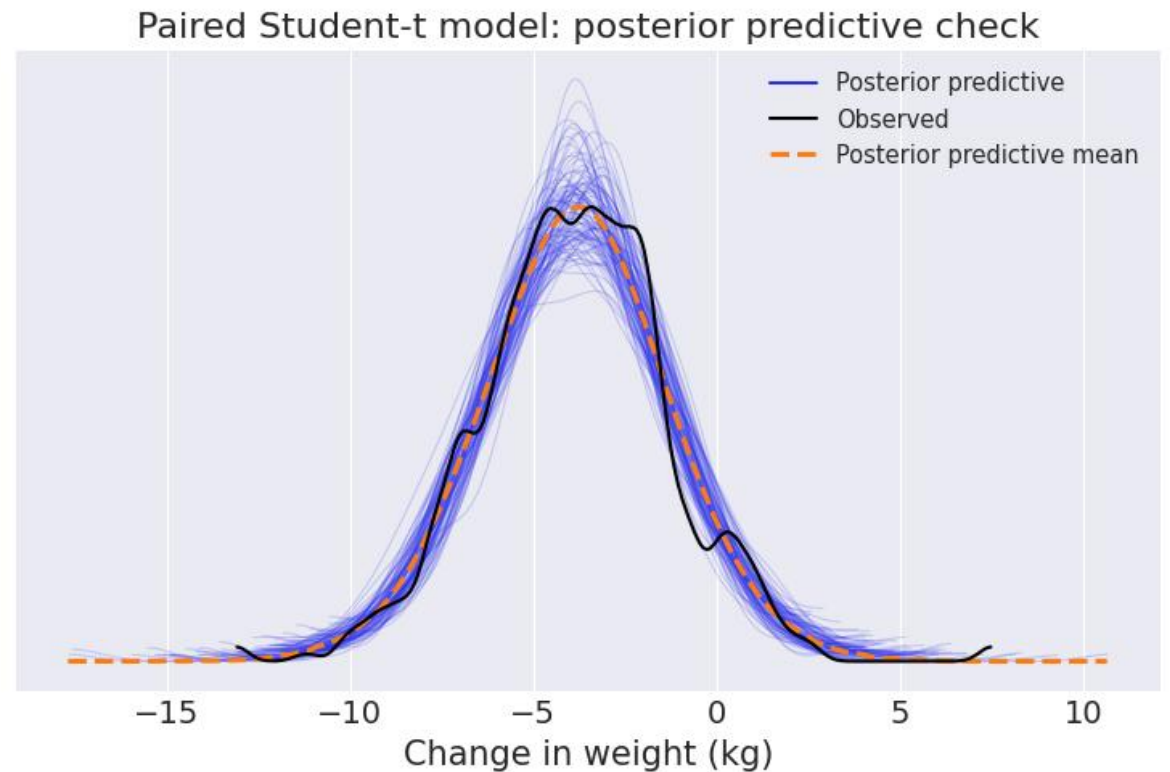
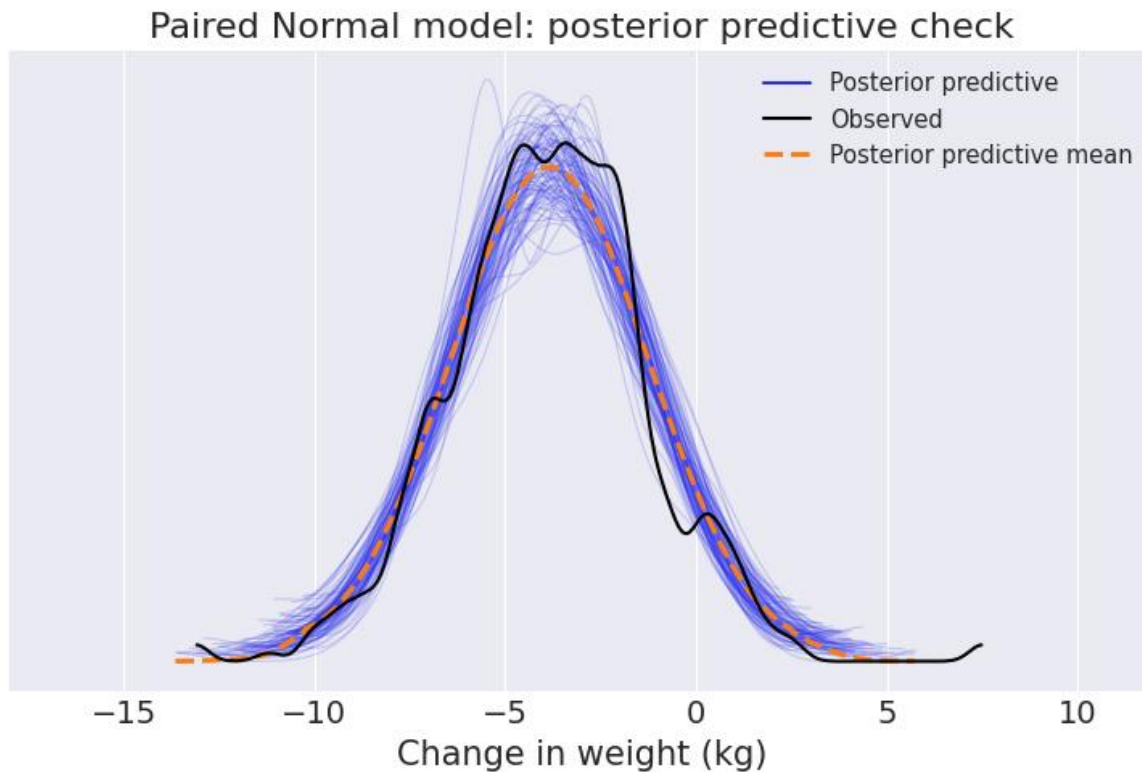
Robust Model

- Posterior Predictive



Robust Model

- Compare Normal and Student-t posterior predictive



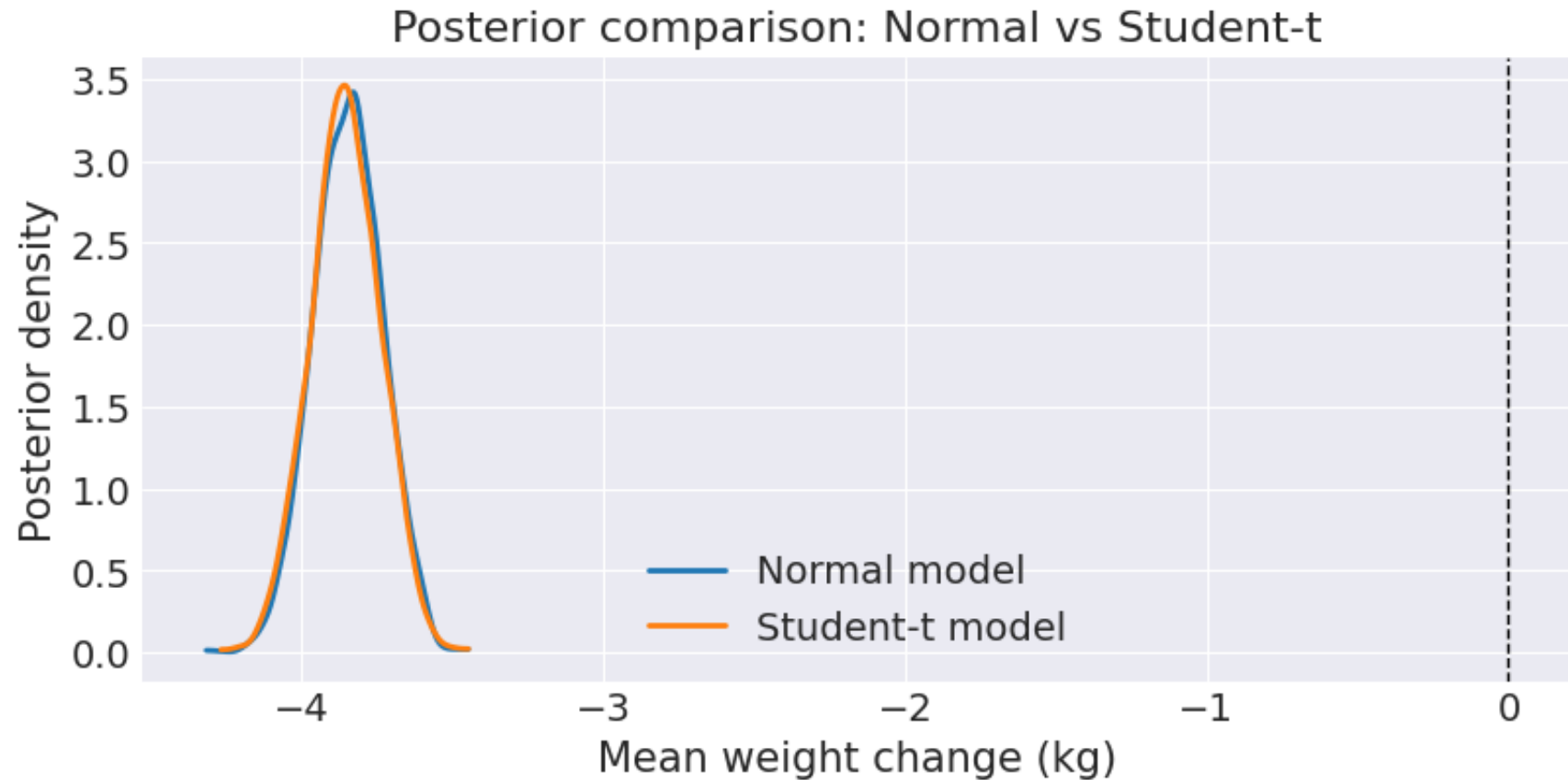
Robust Model

- Compare Normal and Student-t posteriors

	model	quantity	posterior mean	HDI low	HDI high
0	Normal	mu_change	-3.847	-4.064	-3.637
1	Student-t	mu_change	-3.855	-4.067	-3.637
2	Normal	sigma_change	2.536	2.385	2.693
3	Student-t	sigma_change	2.394	2.185	2.589
4	Student-t	nu	26.916	5.865	64.971

Robust Model

- Compare Normal and Student-t posteriors



Unequal variances

- **Shared σ model**

$$y_i \sim \text{Normal}(\mu_{\text{group}[i]}, \sigma)$$

- Assumes:

$$\sigma_A = \sigma_B$$

- **Separate σ model**

$$y_i \sim \text{Normal}(\mu_{\text{group}[i]}, \sigma_{\text{group}[i]})$$

- Allows:

$$\sigma_A \neq \sigma_B$$

Bayesian group model

- The same food diet dataset is used for the independent group comparison.
- We will compare pre-diet weight between male and female participants.
- This is an unpaired comparison because different people belong to each group.

Bayesian group model

- Observed data

	count	mean	std
gender_label			
female	240.0	67.829167	7.548486
male	240.0	79.025833	5.061835



Bayesian group model

- Violin plot:
- Shows the distribution of the data in each group.
- The width represents where more observations are concentrated.
- A wider area means that more participants had weights around that value.
- Helps us see both the center and the spread of each group, not only the average.



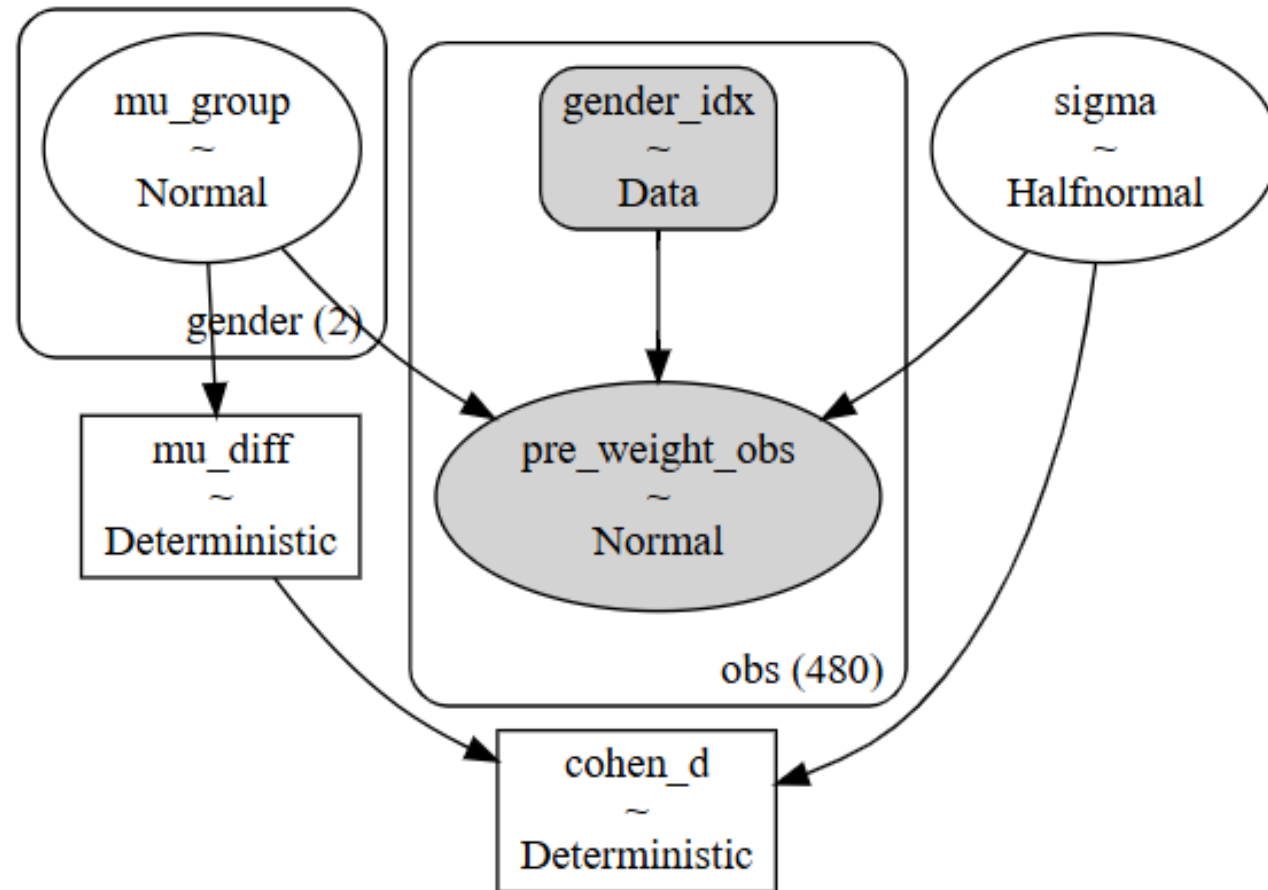
Bayesian group model with **shared** sigma

Model

[illegible]

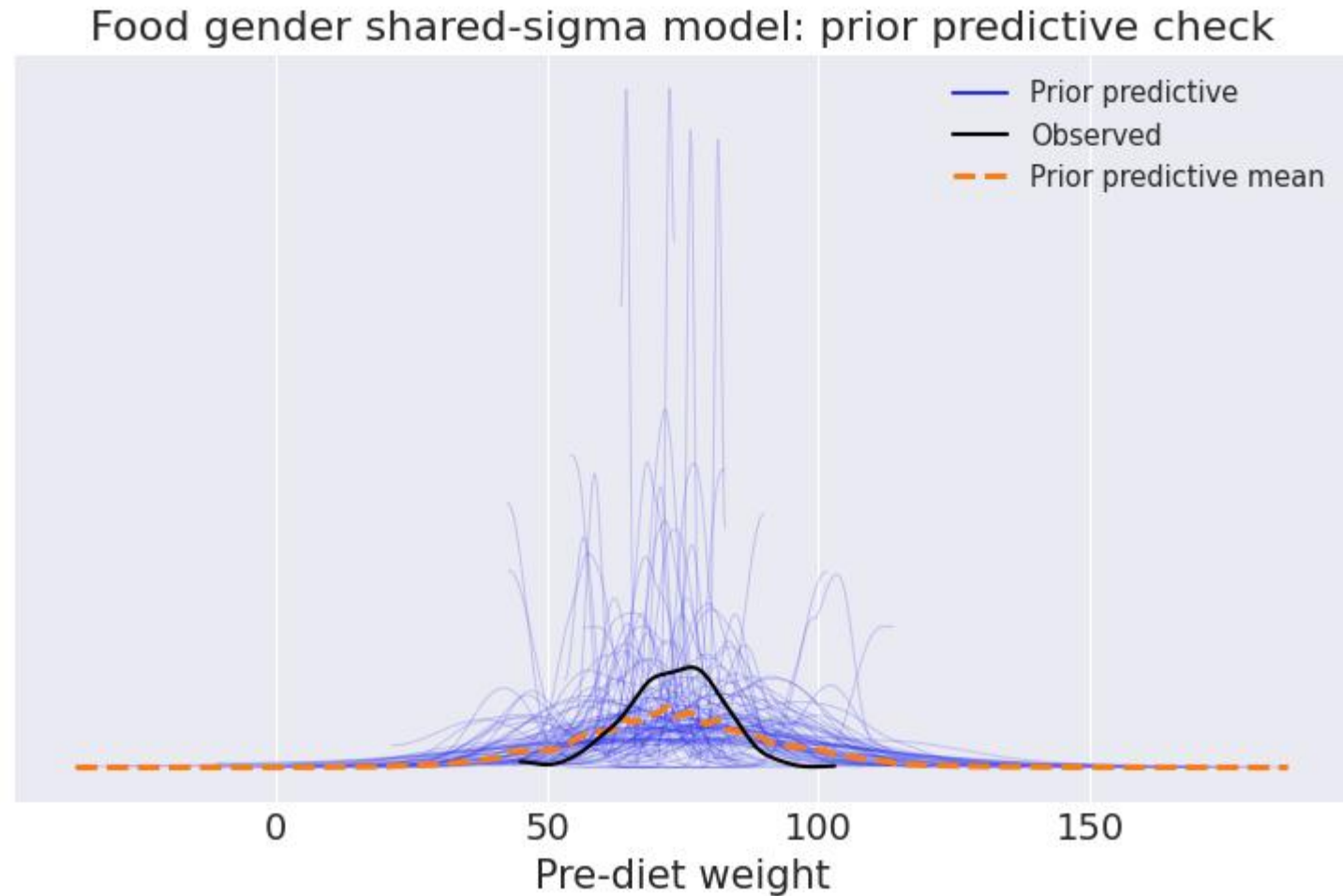
Bayesian group model with **shared** sigma

Model



Bayesian group model with **shared** sigma

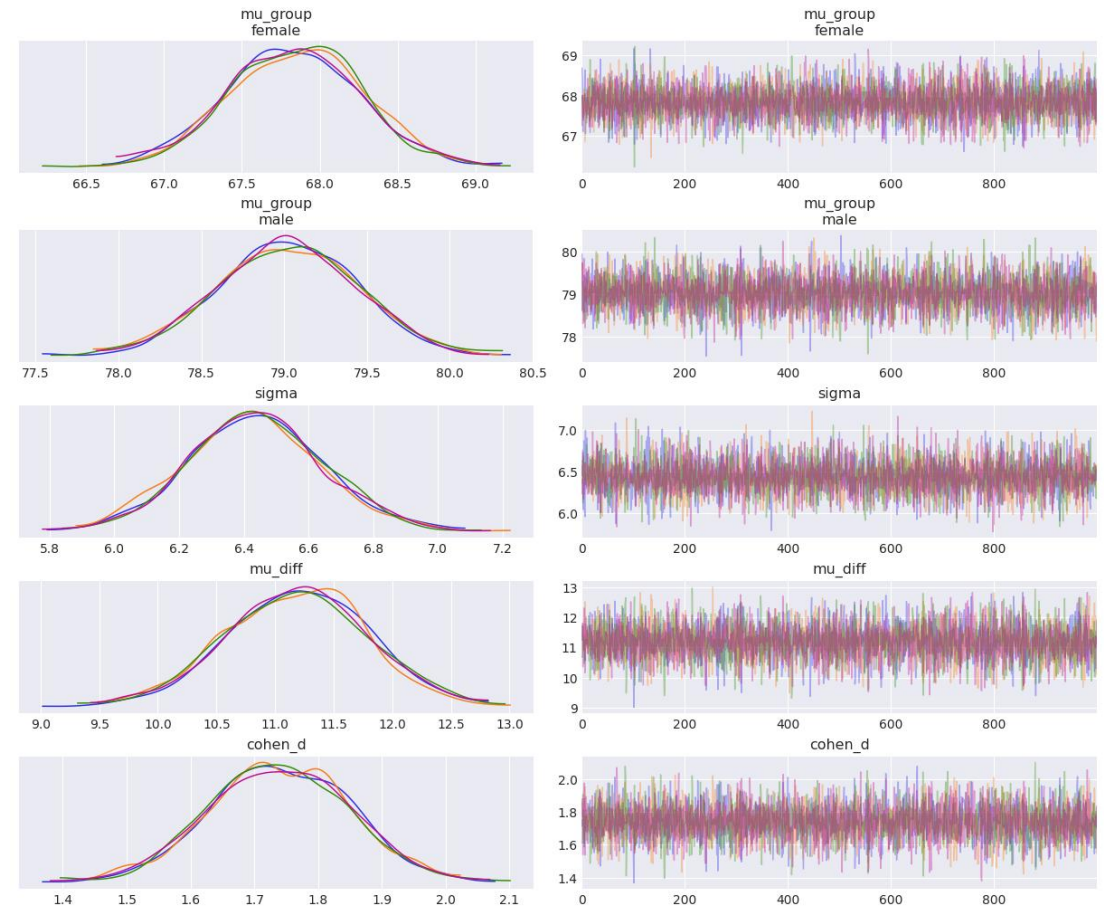
Prior predictive



Bayesian group model with **shared** sigma

Diagnostics

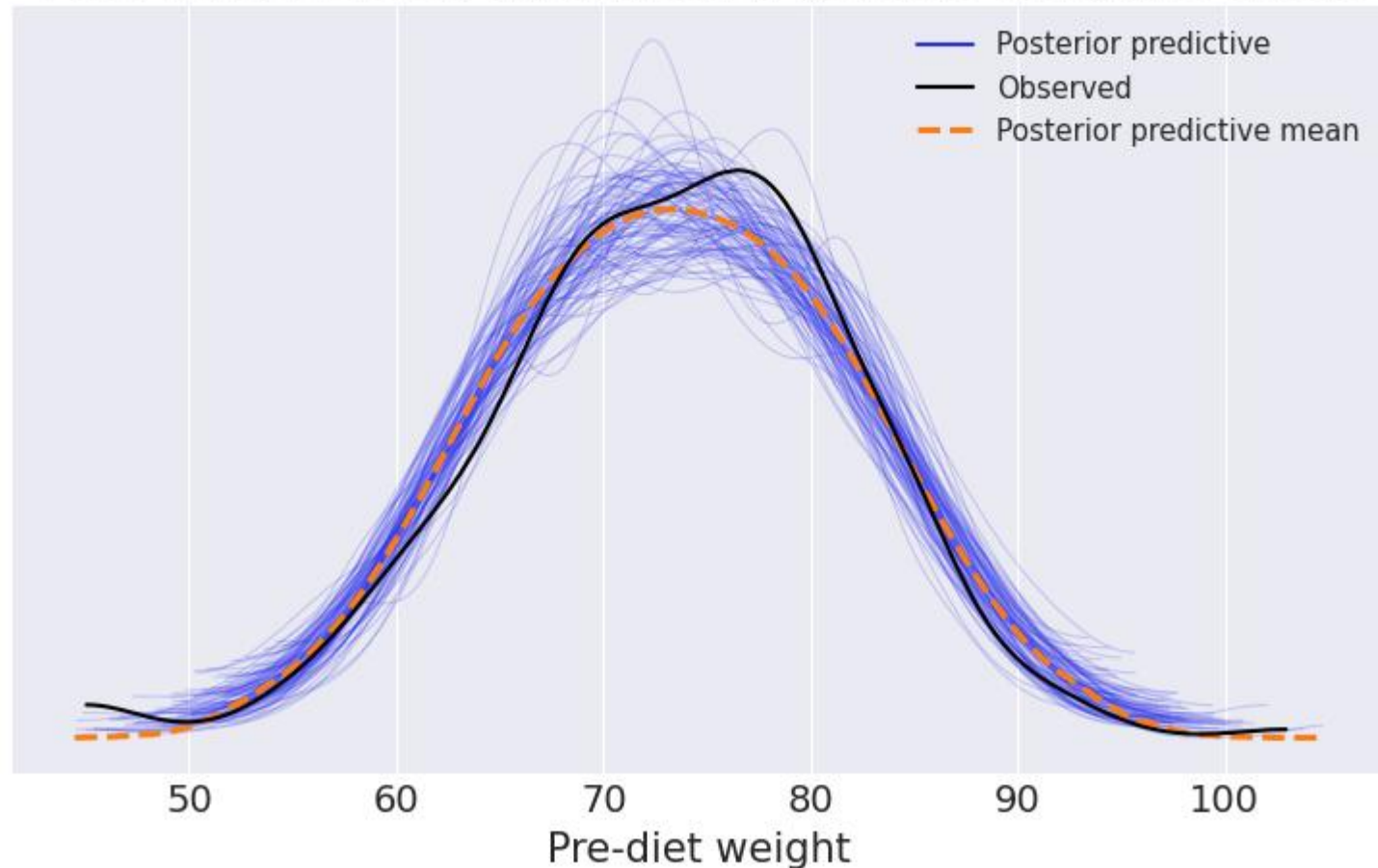
	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_group[female]	0.005	0.006	6888.0	2939.0	1.0
mu_group[male]	0.005	0.006	5920.0	3265.0	1.0
sigma	0.003	0.003	6542.0	3221.0	1.0
mu_diff	0.008	0.009	6051.0	3248.0	1.0
cohen_d	0.001	0.002	6170.0	3292.0	1.0



Bayesian group model with **shared** sigma

Posterior predictive

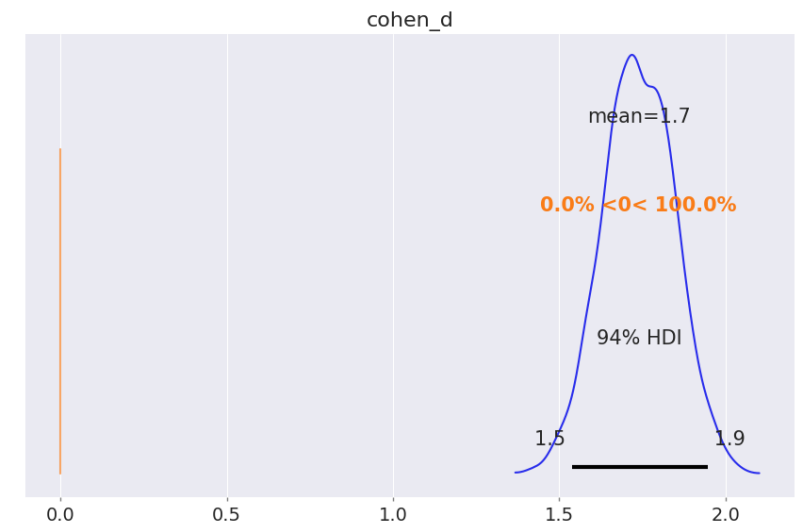
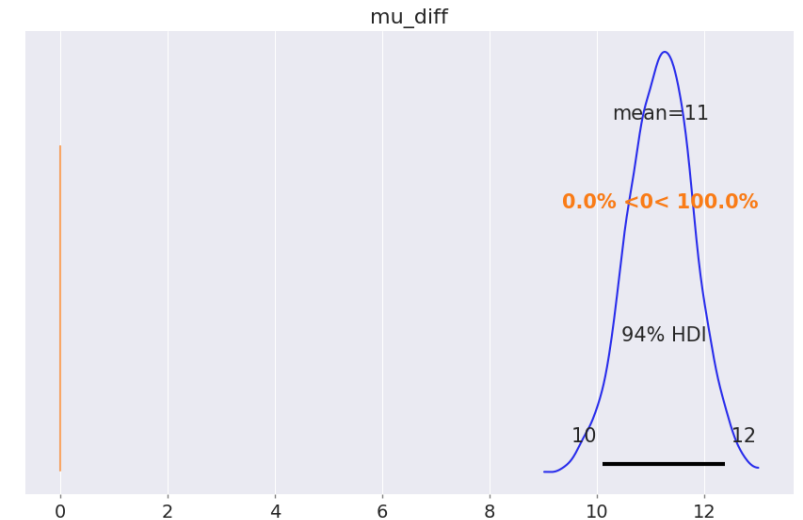
Food gender shared-sigma model: posterior predictive check



Bayesian group model with **shared** sigma

Posterior effect size

	mean	sd	hdi_3%	hdi_97%
mu_group[female]	67.839	0.422	67.031	68.629
mu_group[male]	79.021	0.419	78.222	79.783
sigma	6.436	0.208	6.001	6.796
mu_diff	11.182	0.604	10.097	12.392
cohen_d	1.739	0.109	1.538	1.947

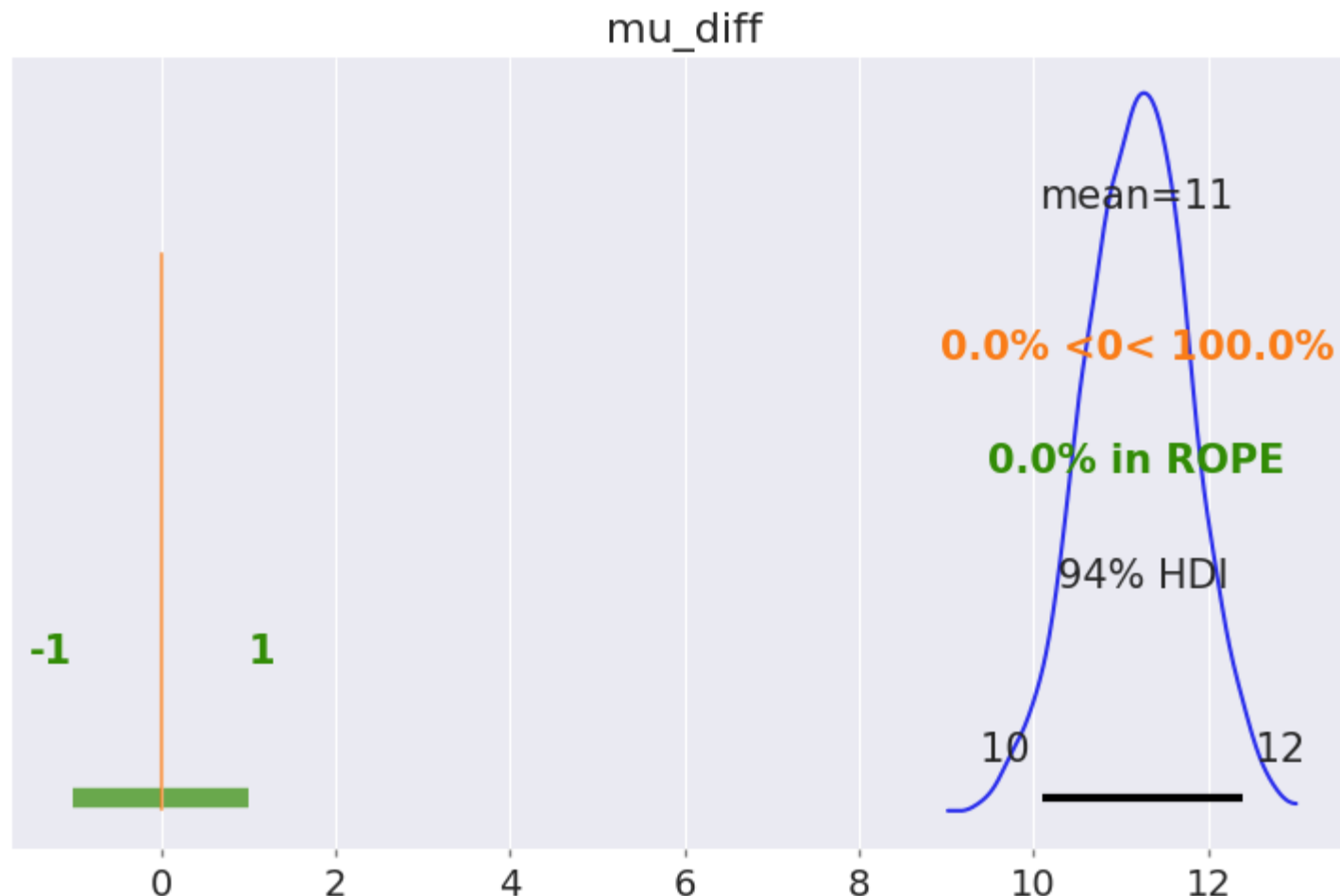


Bayesian group model with **shared** sigma

Posterior

```
rope_pre_weight = (-1, 1)

az.plot_posterior(
    idata_food_gender_shared,
    var_names=["mu_diff"],
    ref_val=0,
    rope=rope_pre_weight,
    hdi_prob=HDI_MASS
)
```



Bayesian group model with **shared** sigma

ROPE\HDI test

	quantity	posterior mean	posterior median	94% HDI low	94% HDI high	Pr(value > 0)	Pr(value < 0)	Pr(value in ROPE)
0	Male - female pre-diet weight	11.182	11.194	10.097	12.392	1.000	0.000	0.000
1	Cohen's d	1.739	1.738	1.538	1.947	1.000	0.000	nan

Bayesian group model with **separate** sigma

- What if the different groups had different sigma's?

Bayesian group model with **separate** sigma

- Model

```
with pm.Model(coords=coords_food_gender) as food_gender_separate_sigma_model:
    # Data
    gender = pm.Data("gender_idx", gender_idx, dims="obs")
    # Priors
    mu_group = pm.Normal("mu_group", mu=72.0, sigma=15.0, dims="gender")
    sigma_group = pm.HalfNormal("sigma_group", sigma=12.0, dims="gender" )

    # Raw effect size
    mu_diff = pm.Deterministic("mu_diff", mu_group[1] - mu_group[0] )

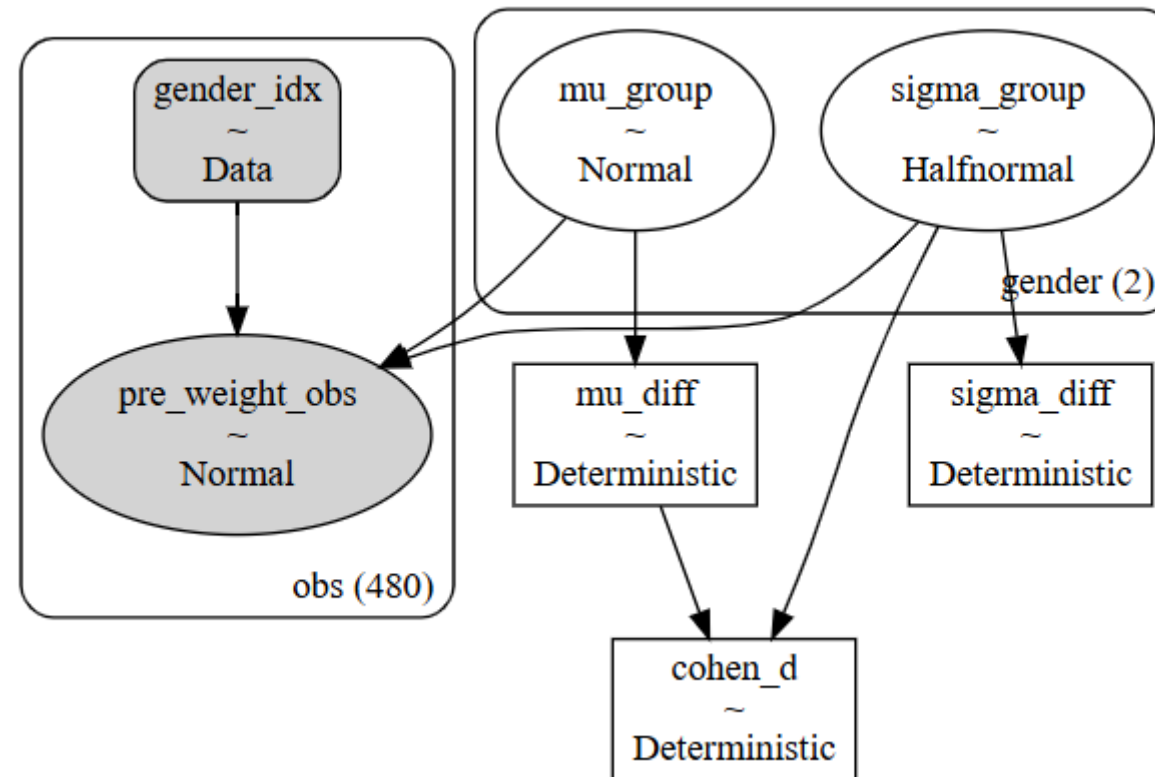
    sigma_diff = pm.Deterministic("sigma_diff", sigma_group[1] - sigma_group[0])

    # Normalized effect size
    cohen_d = pm.Deterministic( "cohen_d", mu_diff / pm.math.sqrt((sigma_group[0]**2
                                                                + sigma_group[1]**2) / 2))

    # Likelihood
    pre_weight_obs = pm.Normal("pre_weight_obs", mu=mu_group[gender],
                               sigma=sigma_group[gender], observed=pre_weight, dims="obs")
```

Bayesian group model with **separate** sigma

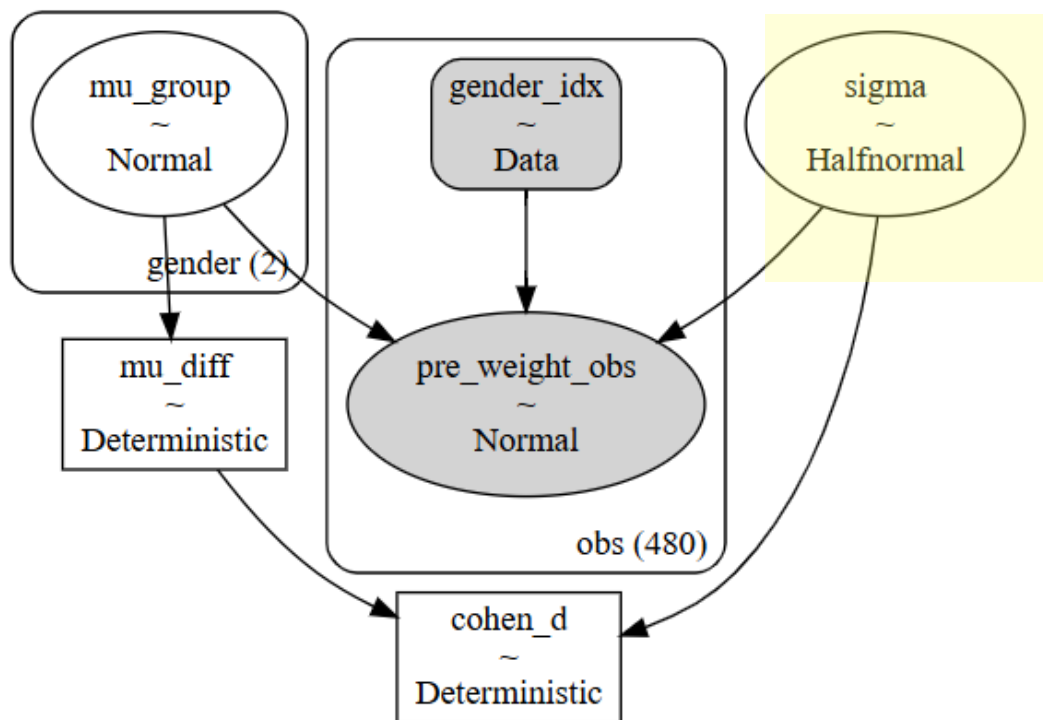
- Model



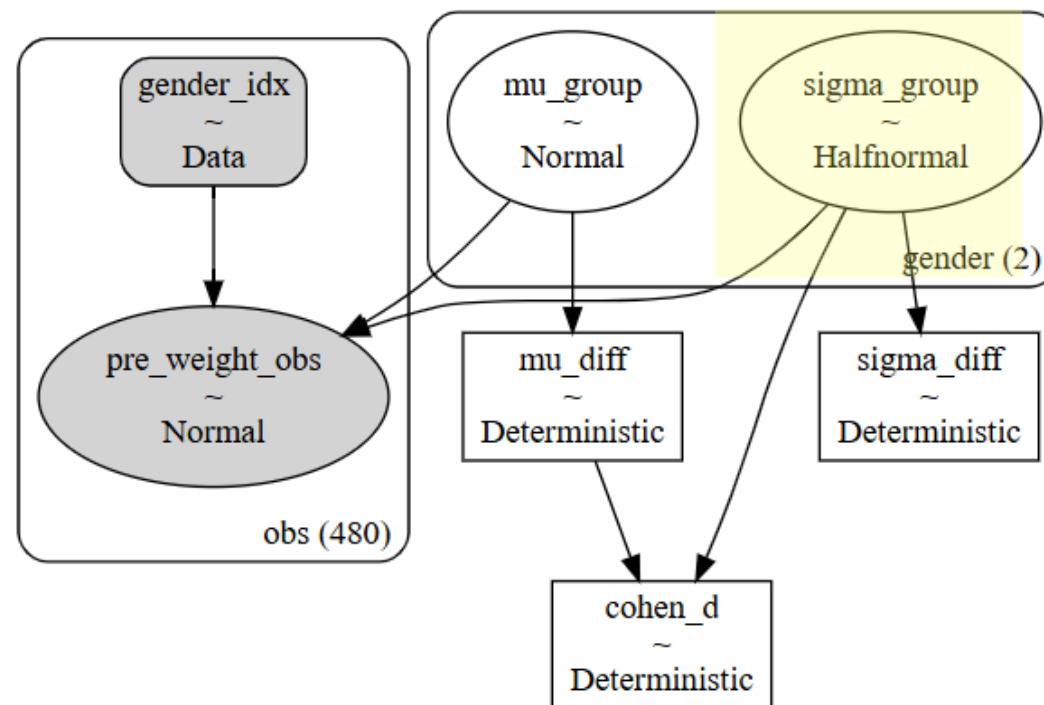
Bayesian group model with **separate** sigma

- Model

grouped sigma



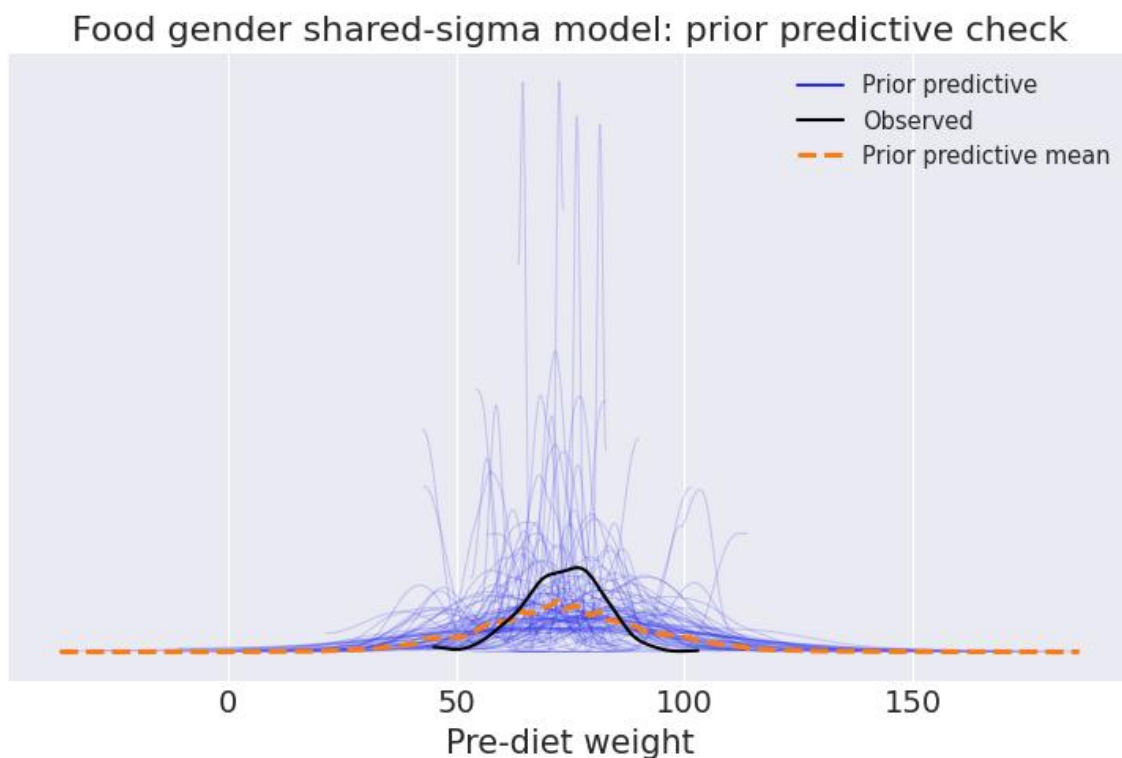
separate sigma



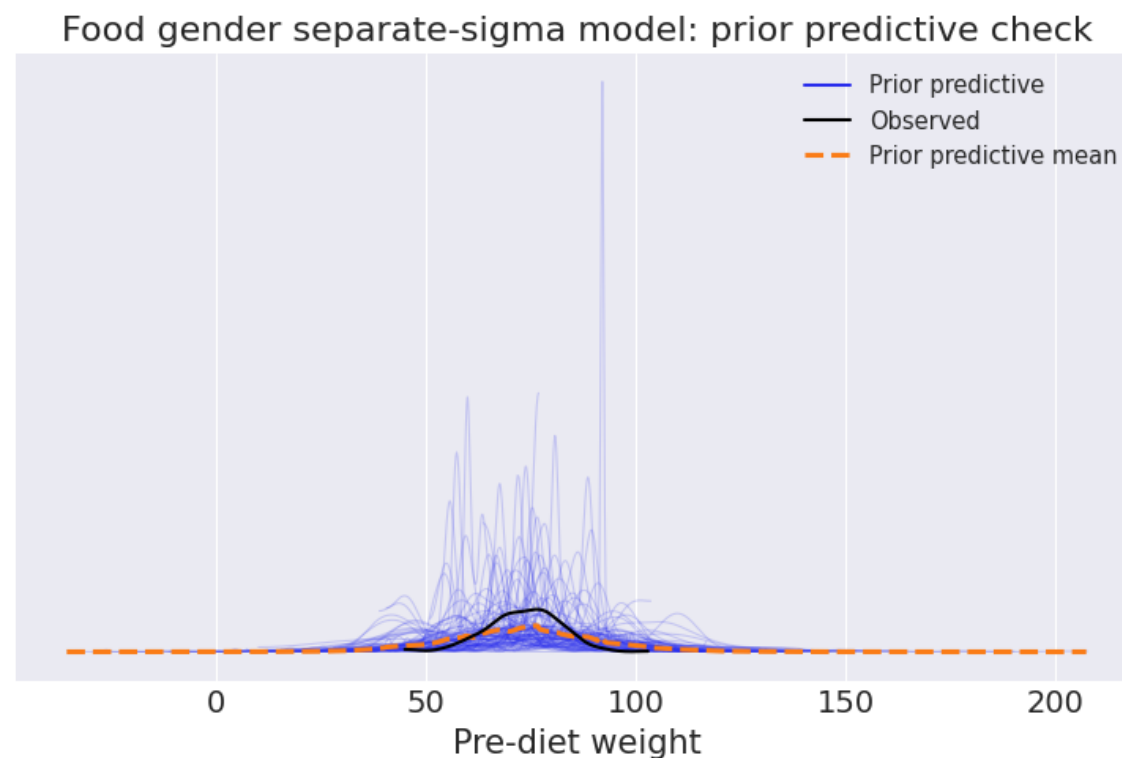
Bayesian group model with **shared** sigma

Posterior predictive

grouped sigma



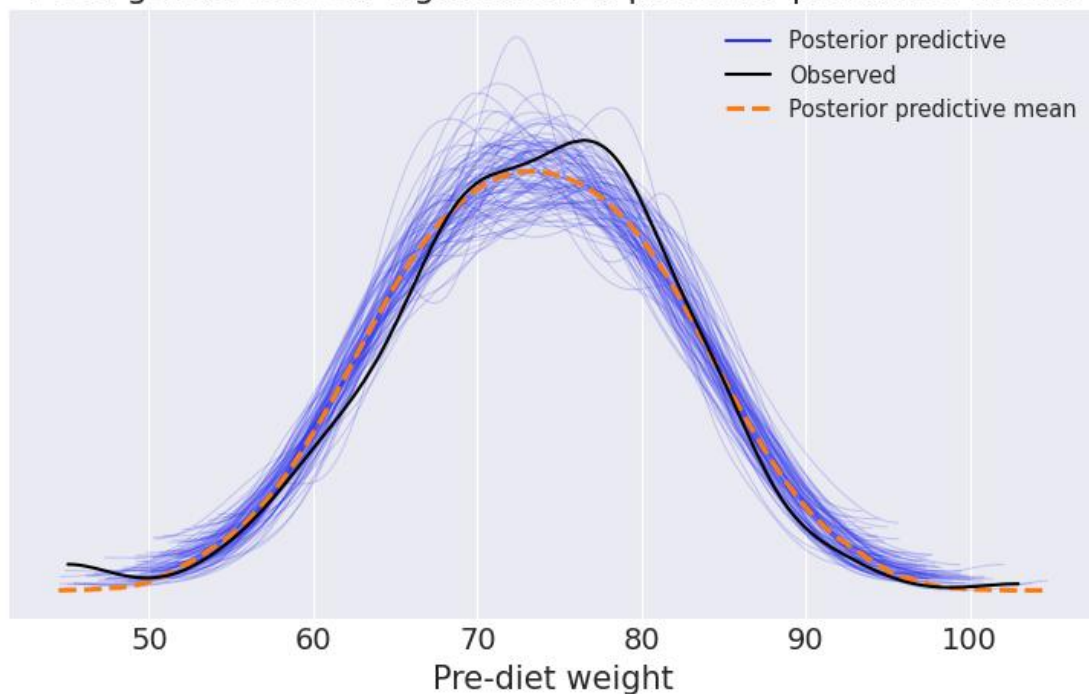
separate sigma



Bayesian group model with **shared** sigma

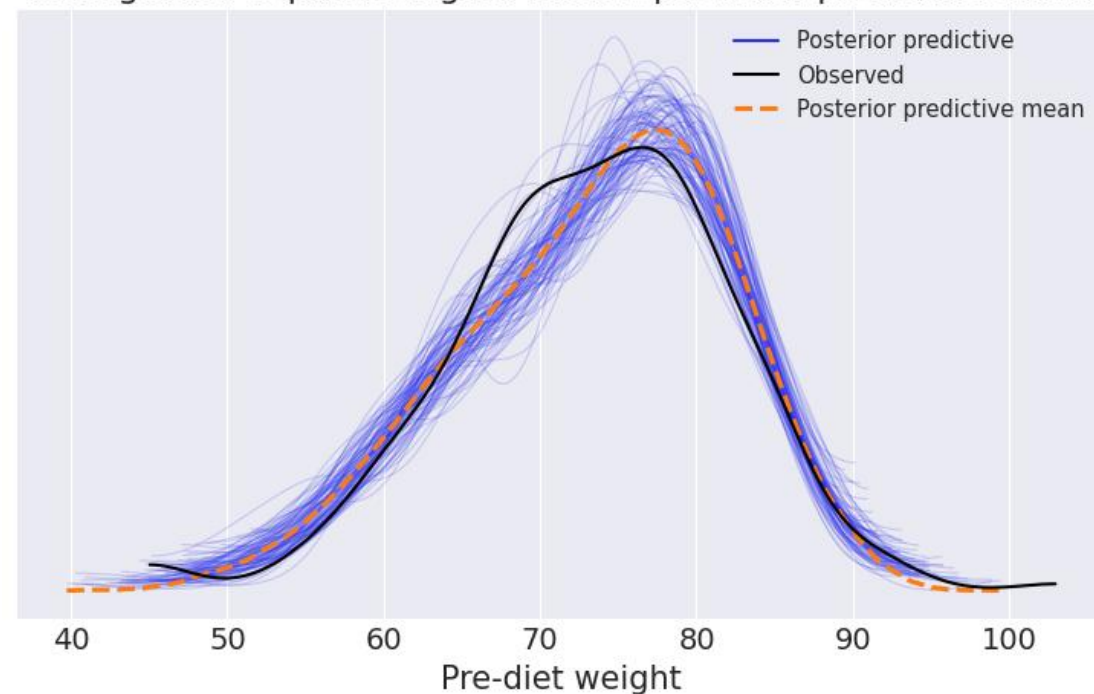
Posterior predictive
grouped sigma

Food gender shared-sigma model: posterior predictive check



separate sigma

Food gender separate-sigma model: posterior predictive check



Bayesian group model with **shared** sigma

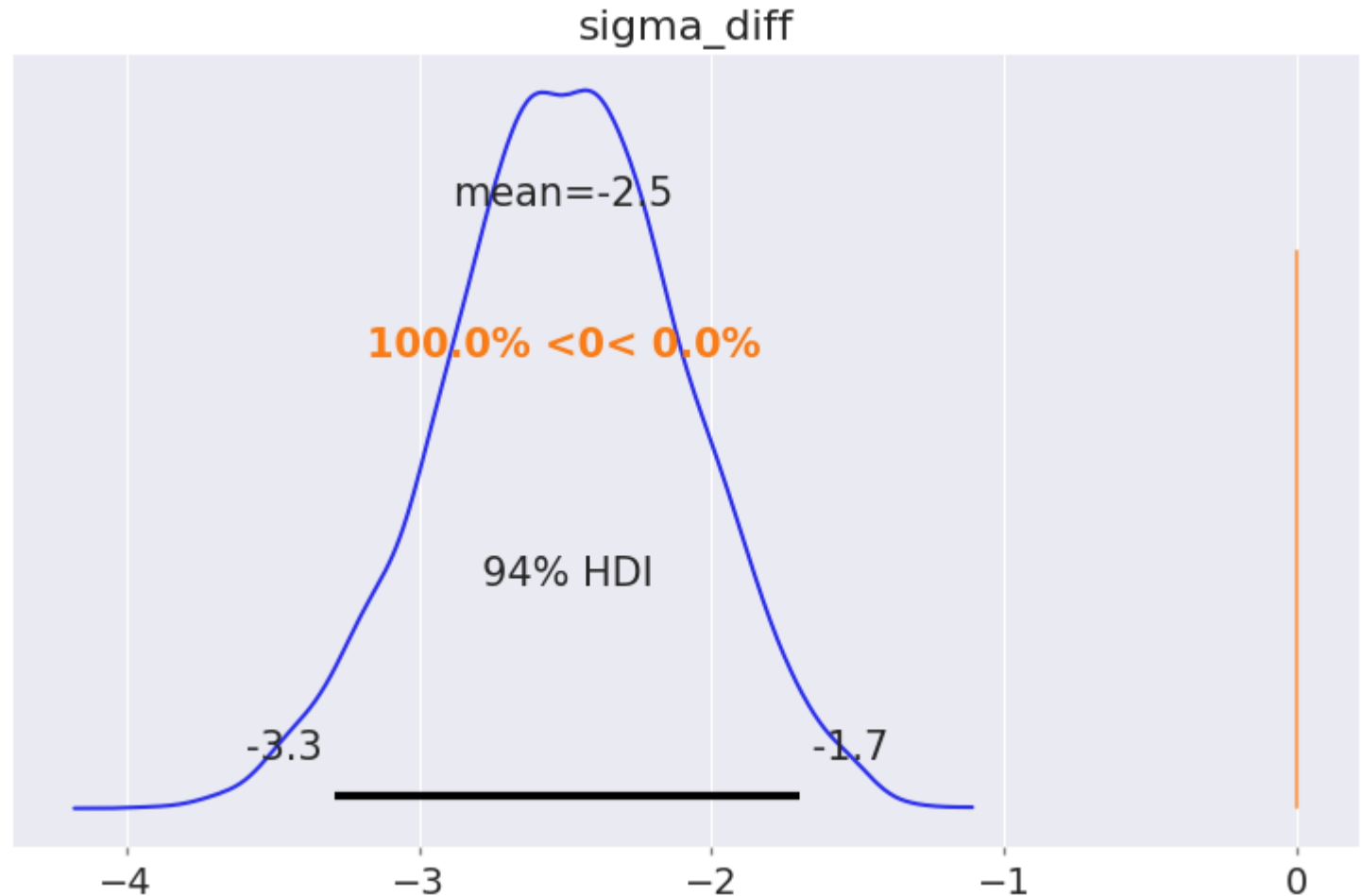
- In this dataset, the posterior mean difference is almost identical in both models.
- The conclusion that males have higher weight is not sensitive to whether we assume equal or unequal variances.

	model	quantity	posterior mean	HDI low	HDI high
0	shared sigma	mu_diff	11.182	10.097	12.392
1	separate sigma	mu_diff	11.193	10.061	12.287
2	separate sigma	sigma_diff: male - female	-2.508	-3.285	-1.701
3	separate sigma	sigma_ratio: male / female	0.671	0.587	0.750

Bayesian group model with **shared** sigma

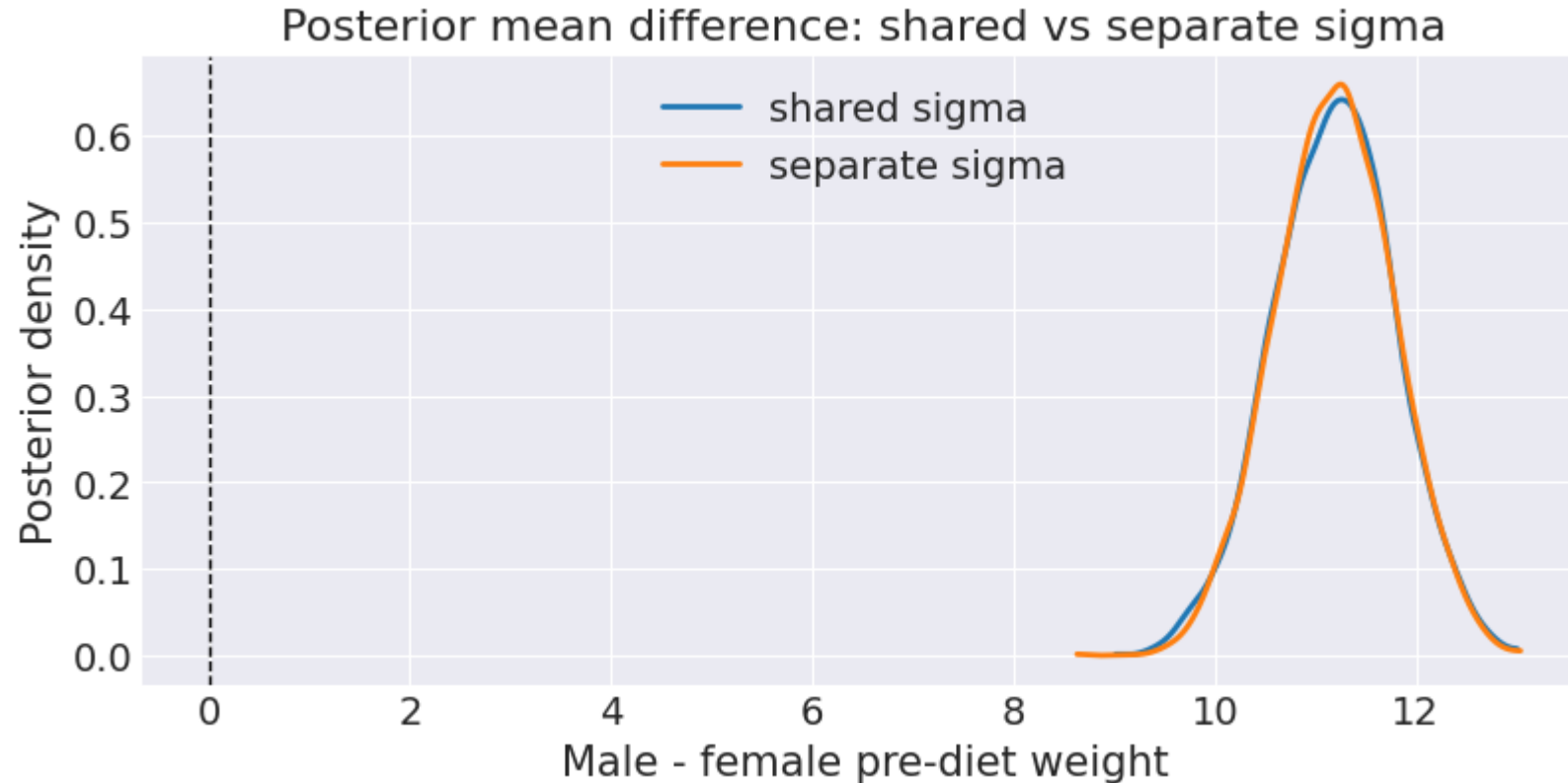
Posterior

- The separate σ model adds information about variability.
- The posterior estimate of $\sigma_{male} - \sigma_{female}$ is negative, and the sigma ratio is under 1 (-0.67), suggesting that females weight may be more variable.
- This affects the spread interpretation, but not the main mean-difference conclusion.



Bayesian group model with **shared** sigma

Posterior comparison



t-Test Overview

Test Type	Common H_0	Description	Example
One-sample t-test	$H_0: \mu = \mu_0$	One continuous variable compared to a reference value	Is average reaction time different from 300 ms?
Paired two-sample t-test	$H_0: \mu_{\text{change}} = 0$	Same subjects measured twice, or matched pairs	Pre/post-treatment, left vs. right
Independent two-sample t-test	$H_0: \mu_A - \mu_B = 0$	Two separate groups	Control vs. experimental group
Welch t-test	$H_0: \mu_A - \mu_B = 0$	Two independent groups, unequal variances allowed	Same comparison, but group spreads may differ
Mann–Whitney	Distributions differ between groups	Two independent groups	Rank-based independent-group alternative
Wilcoxon	Median paired change =0	Paired / repeated measurements	Rank-based paired alternative

Independent t-Test

- Observed data

	count	mean	std
gender_label			
female	240.0	67.829167	7.548486
male	240.0	79.025833	5.061835



Independent t-Test

- In independent t-test we assume independent measurements.
- Here, we compare **pre-diet weight** between males and females:

$$H_0: \mu_{male} - \mu_{female} = 0$$

$$H_1: \mu_{male} - \mu_{female} \neq 0$$

Independent t-Test

- The test statistic is:

$$t = \frac{\bar{x}_{male} - \bar{x}_{female}}{s_p \sqrt{\frac{1}{n_{male}} + \frac{1}{n_{female}}}}$$

- Where:

- $\bar{x}_{male}, \bar{x}_{female}$ are the sample means,
- n_{male}, n_{female} are the sample sizes,
- s_{male}, s_{female} are the sample standard deviations,
- s_p is the pooled standard deviation

$$s_p = \sqrt{\frac{(n_{male} - 1)s_{male}^2 + (n_{female} - 1)s_{female}^2}{n_{male} + n_{female} - 2}}$$

- Under the null hypothesis:

$$t \sim t_{n_{male} + n_{female} - 2}$$

- If the p-value is smaller than α , we reject H_0

Independent t-Test

```
female_weight = food_group.loc[food_group["gender_label"] == "female", "pre.weight"].to_numpy()
male_weight = food_group.loc[food_group["gender_label"] == "male", "pre.weight"].to_numpy()
```

```
pooled_t_result = stats.ttest_ind(
    male_weight,
    female_weight,
    equal_var=True
)
```

```
mean_diff_frequentist = male_weight.mean() - female_weight.mean()
```

```
n_male = len(male_weight)
n_female = len(female_weight)
```

```
sd_male = male_weight.std(ddof=1)
sd_female = female_weight.std(ddof=1)
```

```
pooled_sd = np.sqrt(
    ((n_male - 1) * sd_male**2 + (n_female - 1) * sd_female**2)
    / (n_male + n_female - 2)
)
```

```
cohens_d_independent = mean_diff_frequentist / pooled_sd
```

```
se_pooled = pooled_sd * np.sqrt(1 / n_male + 1 / n_female)
df_pooled = n_male + n_female - 2
t_critical_pooled = stats.t.ppf(1 - SIGNIFICANCE_ALPHA / 2, df=df_pooled)
pooled_ci_low = mean_diff_frequentist - t_critical_pooled * se_pooled
pooled_ci_high = mean_diff_frequentist + t_critical_pooled * se_pooled
```

scipy.stats.
ttest_ind

```
ttest_ind(a, b, *, axis=0, equal_var=True, nan_policy='propagate',
permutations=None, random_state=None, alternative='two-sided', trim=0,
method=None, keepdims=False)
```

[\[source\]](#)

Calculate the T-test for the means of two *independent* samples of scores.

This is a test for the null hypothesis that 2 independent samples have identical average (expected) values. This test assumes that the populations have identical variances by default.

Independent t-Test

	test	statistic	df	p-value	mean difference	CI low	CI high	Cohen's d
0	Pooled independent t-test	19.085	478.00	8.826e-61	11.197	10.044	12.349	1.742

- **t-statistic:** 19.085
- **p-Value :** $8.826 \cdot 10^{-61}$ (< 0.05) $\Rightarrow$ low value, indicating strong statistical significance
- **CI:** [10.04, 12.34]
- **Cohen's d:** 1.742 (large effect size)

Independent t-Test

- **Reporting**
- Males had higher pre-diet weight than females. The difference was statistically significant, $t(478) = 19.085$, $p < 0.001$. The estimated mean difference was 11.197 kg, with a 95% confidence interval of [10.04, 12.34]. The standardized effect size was large, $\text{Cohen's } d = 1.742$.

Independent t-Test

- **Welch t-test**
- Used for the same comparison as the independent two-sample t-test:

$$H_0 : \mu_A - \mu_B = 0$$

- but it does not assume equal variances between the two groups. :

$$\sigma_A^2 \neq \sigma_B^2$$

	test	statistic	df	p-value	mean difference	CI low	CI high	Cohen's d
0	Pooled independent t-test	19.085	478.00	8.826e-61	11.197	10.044	12.349	1.742
1	Welch t-test	19.085	417.79	7.584e-59	11.197	10.043	12.350	1.742

Tutorial 6: Main ideas

- **Continuous data** require continuous likelihoods.
- **The Normal model** describes data using center (μ) and spread (σ)
- **Addition to the Bayesian workflow:**
 - Bayesian p-values as model-fit checks
- **Paired data:**
 - Analyze within-subject change scores.
- **Group data:**
 - Use group-specific means, and decide whether σ should be shared or group-specific.
- **Student-t likelihood:**
 - A robust alternative when data have heavy tails or outliers.
- **Frequentist t-tests:**
 - Answer questions through null sampling distributions and p-values rather than posterior uncertainty.